Substituting Reference σ-Profiles Degrades Solubility Prediction; on Activity Data, Bounding a Deployed COSMO-SAC Against Its Inputs Returns a Map, Not a Verdict

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Abstract

Physics-informed models route predictions through interpretable physical quantities, expecting accurate references to help. We measure that for solid–liquid solubility: a graph neural network predicts charge-density profiles a quantum-chemical database tabulates; fixed COSMO-SAC and a crystal term turn two into a solubility. Grounding names two operations: supervising the profile against that database during training lowers mean absolute error, over three seeds, not certified leak-free; substituting the reference for the learned profile raises it, at inference and—by less, at one seed—during training, where it co-adapts. Neither resolves against the gain; only the signs carry. On infinite-dilution activity data another such database bounds a deployed thermodynamic model's error from below, its inputs' from above. The input bound barely moves—an estimator property, not chemistry—while the error does: the output is a map, most strata unboundable. One admissibility rule leaves one row set standing, glycol-ether solvents, where that model's error exceeds its inputs' by a margin itself only a lower bound—and no other stratum, neither set whole, nothing on solubility. It is the survivor of a search across every stratum, not an established effect: chemistry-blind relabelling certifies some row set in most draws, reaching that margin in a tenth. A pre-declared out-of-sample test returned *not testable*: the one independent database holds no glycol-ether solvent with an organic solute. Trained through this closure, the learned profile departs from its reference along a shared direction, on separate three-seed runs. A Hammett pK_a closure reverses the substitution's sign where its constants suffice.

1 Introduction

How much of a solid dissolves in a given liquid decides whether a drug can be formulated as a tablet or an injection, which solvent a crystallisation is run in, and how a product is purified. It must be measured one solute, one solvent and one temperature at a time, so most industrially relevant combinations have never been determined, and independent determinations of the same system disagree by enough to floor what any predictor can achieve. Dissolving a solid also trades the cost of breaking up a crystal against the cost of mixing the freed molecules into the solvent, so a predictor has to be right about two unrelated kinds of chemistry at once, on molecules unlike any it was shown.

Predictors divide by how much physical structure they impose. Direct models regress solubility from descriptors or from a representation learned off the molecular graph; FastSolv[1] does so with a deep neural network and leads on extrapolation to new solutes. They are the more accurate there and they are opaque: the prediction arrives as a single number, with no account of which part of the chemistry it captured. Physics-informed models instead predict the thermodynamic quantities the physical picture names—an activity coefficient for the solute–solvent interaction, a melting temperature and heat of fusion for the crystal—and combine them through a fixed equation, as SolProp[2] does with a thermodynamic cycle of separately learned components. What that buys is decomposability: each part is a quantity chemists tabulate on its own, so it can be supervised against reference data a direct regressor has no way to use. On accuracy the black box leads and the physics-informed cycle trails it. Learned surrogates for the activity coefficient have advanced quickly, and split the same way. Winter et al.[3] predict $\ln \gamma_2^\infty$ from SMILES with a language model and Sanchez-Medina et al.[4] embed a Gibbs–Helmholtz constraint in a graph network for its temperature dependence, both making the activity coefficient the regression target; a second line instead keeps a mechanistic model and learns its input. COSMO-SAC[5] is a segment-activity reformulation of Klamt's COSMO-RS, and the σ-profile it consumes—screening-charge density over a molecule's cavity—is that framework's construct, tabulated from quantum chemistry.[6] Predicting it from structure removes that calculation from the pipeline; TeNNet-SAC[7] does so while also learning the segment-activity kernel, and gains from truer profiles.

When a fixed physical structure improves or degrades a learned model is the broader question of physics-informed machine learning,[8] whose dominant forms impose structure as a differentiable object: a governing equation enforced through the loss,[9] gray-box hybrids composing learned components with fixed mechanistic maps,[10] and operator learning with a trainable correction to a misspecified equation.[11] The prevailing expectation is that more physics improves generalisation, yet the same structure has documented failure modes in which it makes optimisation or accuracy worse:[12] in neural PDE solvers, a learnable term added to a fixed discretised operator absorbs truncation artifacts rather than physics, so the apparent interpretability is illusory[13]—though that error source is numerical and has no external reference. The same concern appears in concept-bottleneck models as *concept leakage*, a learned concept silently encoding information beyond its nominal meaning,[14–16] and recent work traces one route to it to the downstream model: a misspecified head forces the encoder to deform the concept into carrying a dependence the head cannot represent,[17] and overwriting a frozen head's leaked concepts with ground truth can then degrade accuracy.[18] The gap between a fixed physical map and reality is the model-discrepancy problem of Kennedy and O'Hagan[19] and its calibration critique;[20] four further literatures—forecast decomposition, quasi-maximum likelihood, identifiability, and the validity of composed pipelines—supply pieces used below.

The design rests on an assumption rarely stated because it is taken for self-evident: a model supplied with physical inputs closer to their true values predicts better. For one class of intermediate that can be tested outright, because a reference value is tabulated and can be handed to the model in place of the one it computed. It needs testing, because when a predictor of this shape is inaccurate either the fixed physical model is wrong or the learned quantities feeding it are uninformative, and the two call for opposite remedies. What this literature lacks is a way to *measure*, rather than diagnose, which of the two limits a composed predictor's accuracy, and whether the learned intermediate has stopped reporting the quantity it is named after; in the concept-bottleneck case no external reference exists against which either question could be settled. This work supplies one—a reference σ-profile for solubility, tabulated substituent constants for a second chemical domain—and holds the mechanistic model fixed, so the tabulated value stays available as an external reference for the learned one.

Here a graph neural network predicts the pair of charge-density distributions COSMO-SAC consumes, in place of the quantum-chemical calculation that ordinarily supplies them. The tabulated counterpart can enter at either of two points: as a target the network is trained toward, or as a replacement for what the network computed at the moment of prediction (Fig. 1). The thermodynamic model is differentiable, so both routes are open on a single trained network, and the tabulated values are put to a third use, dividing the fixed model's error between the map and the inputs handed to it. Only one route helps. Whether the fixed map is what makes the other one fail is a separate question and is asked on separate data, because solubility data cannot say which component is at fault, one measurement constraining the pair jointly; activity coefficients can, because both molecules carry a tabulated distribution and the two error terms separate. Where they separate, an ordering decides the remedy: a map whose own error stands above what its inputs fail to carry does not become more accurate when handed a better input, but more exposed, the accurate input no longer masking the bias the map introduces. That ordering is a property of the chemistry the map is asked about, so the instrument returns a map over chemical strata—one answer per chemically defined set of rows—rather than a verdict. Two consequences follow. A learned intermediate trained through a misspecified map cannot be read as the physical quantity it is named after: the predicted distribution drifts away from the tabulated one along a direction shared across molecules—measured on a separate three-seed set of runs, not the checkpoints whose paradox it is offered to explain. And substituting a reference value for it at prediction time is not a safe improvement but a change of regime; here it degrades prediction at every seed. Neither consequence is peculiar to charge distributions: tabulated substituent constants behave the same way inside a fixed Hammett relation for acid dissociation, improving the records the tabulated constants describe completely—every substituted position meta or para with a constant in the table—and degrading those carrying an ortho or an untabulated substituent, a division that runs across all four reaction series and not between them.

2 Computational methods

2.1 The composed predictor and its error split

For a solute s in solvent v at temperature T, the target is $y = \ln x_2$, the log saturated mole fraction. Dissolving a solid trades breaking the crystal lattice (Φ) against mixing the freed molecules into the solvent ($\ln \gamma_2$), and solid–liquid equilibrium (SLE) sets the log-solubility to their negative sum:

$$\ln x_2 = \underbrace{-\frac{\Delta H_{\text{fus}}}{R}\Big(\tfrac{1}{T} - \tfrac{1}{T_m}\Big)}_{\Phi(T)\ \text{(crystal)}} - \underbrace{\ln \gamma_2(x_2, T)}_{\text{activity}}, \quad (1)$$

with the constant-ΔC_p correction omitted for clarity. The activity term is produced by a *differentiable closure* g: a graph encoder predicts a σ-profile—the distribution of screening charge density over a molecule, the histogram a COSMO model consumes—for each molecule, and COSMO-SAC,[5] a segment-activity reformulation of COSMO-RS, maps the pair of profiles to $\ln \gamma_2$. The segment-interaction energy inside that map—the electrostatic-misfit and hydrogen-bonding terms setting what two surface segments cost each other—is the closure's *kernel*, the part the published revisions rewrite, so a closure variant is named for the year of its kernel: the deployed *2002 kernel*, and the *2010 kernel* that types segments as hydrogen-bond donors or acceptors instead of treating hydrogen bonding with one parameter. Write $z^\star$ for the *reference* physical latent, the σ-profile tabulated in the VT-2005 database,[6] and $\hat{z}$ for the head's own prediction. Feeding $z^\star$ in place of $\hat{z}$ (a σ-oracle) isolates the closure from encoder error: if the closure were well specified, $g(z^\star)$ should be the best available activity estimate. Each trained configuration, and each closure variant evaluated, is an *arm*; a *Direct-GNN* control arm drops the closure and emits $\ln x_2$ directly, and the two were tuned sepa-

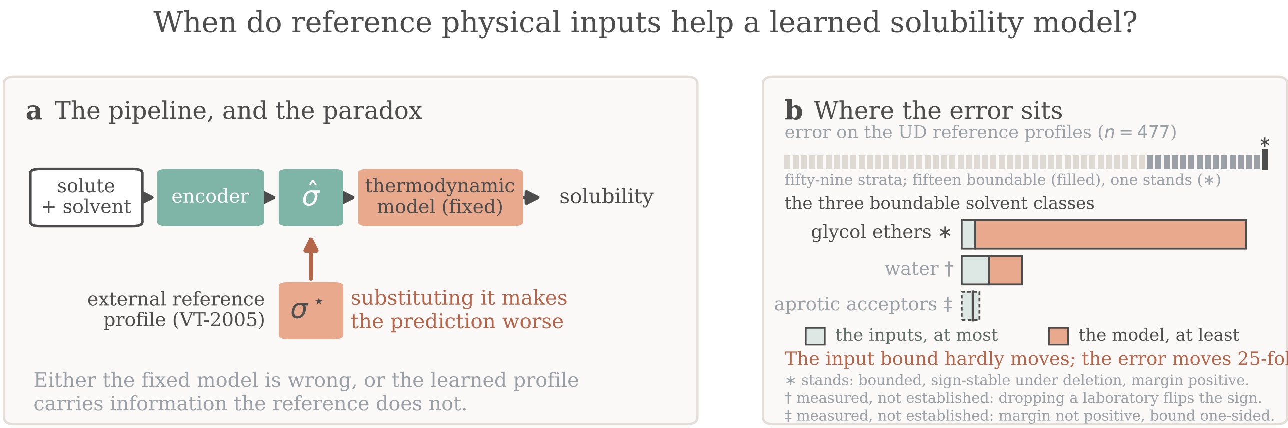

Figure 1: **Overview.** **(a)** A learned encoder predicts a charge-density profile $\hat{\sigma}$ that a fixed thermodynamic model turns into a solubility; the arrow marks the evaluation-time substitution of the reference $\sigma^\star$, which makes it worse. **(b)** The counterpart error on infinite-dilution activity data: the strip is every stratum searched, the bars the three boundable solvent classes; only the starred class stands, each mark carrying its own reason, and each block is a one-sided bound rather than an interval. Training-time supervision is not drawn.

rately (§2.4).

Everything below is stated for a general *composed predictor* $\hat{y} = g(h(x))$ in which a fixed closure g consumes a learned physical intermediate $z = h(x)$, against a matched free predictor that shares the encoder and drops g; solubility is one instance and the pKa closure of §3.6 a second.

The crystal/activity split of Eq. (1) is not identifiable from solubility alone. At infinite dilution with $\Delta C_p = 0$ the observable is affine in $1/T$, and the activity intercept and slope already span that plane, so the profiled Fisher information for ΔH_{fus} is exactly zero and the indeterminacy set is two-dimensional; at finite dilution the composition factor $(1 - x_2)^2$ leaves an information of order x_2^2. The deficiency closes only under a rank-completing external constraint—a direct crystal label, or a fixed activity slope (Lemma S1, §S2). That is why an under-determined factor needs external single-component data and why compensation between the factors is expected; it says nothing by itself about whether grounding helps.

2.1.1 The closure–insufficiency decomposition

Let m be the closure's target—here the experimental infinite-dilution activity coefficient (IDAC) $\ln \gamma_2^\infty$, in §3.6 the measured pK_a—and $g(z^\star)$ the closure evaluated on the reference latent.

Lemma 1 (Exact closure–insufficiency decomposition and its one-sided bounds)**.** *For a fixed (non-fitted) closure g, with $z^\star$ denoting the closure's* entire *argument (the profile pair together with the temperature wherever the closure reads one),*

$$\underbrace{\mathbb{E}\big[(m - g(z^\star))^2\big]}_{\textit{ref.-input MSE}} = \underbrace{\mathbb{E}\big[\mathrm{Var}(m \mid z^\star)\big]}_{B_{\text{insuff}}\ (\textit{irreducible})} + \underbrace{\mathbb{E}\big[(\mathbb{E}[m \mid z^\star] - g(z^\star))^2\big]}_{B_{\text{closure}}\ (\textit{decoder bias})}, \tag{2}$$

the cross term vanishing identically. Moreover $B_{\text{closure}} \geq (\mathbb{E}[m] - \mathbb{E}[g])^2$, and $B_{\text{insuff}} \leq$ $\mathbb{E}[\mathrm{Var}(m \mid \mathrm{bin}(g(z^\star)))]$ for any binning of $g(z^\star)$.

Proof. (a) The identity, and the cross-term condition. $g(z^\star)$ is $\sigma(z^\star)$-measurable by construction, since $z^\star$ is the closure's whole argument. Split the residual as $m - g(z^\star) = [m - \mathbb{E}(m \mid z^\star)] + [\mathbb{E}(m \mid z^\star) - g(z^\star)]$, square and take expectations; the squared terms are B_{insuff} and B_{closure}, and the cross term vanishes by the tower rule, $\mathbb{E}[(m-\mathbb{E}(m \mid z^\star))(\mathbb{E}(m \mid z^\star)-g(z^\star))] = \mathbb{E}[(\mathbb{E}(m \mid z^\star)-g(z^\star))\,\mathbb{E}(m-\mathbb{E}(m \mid z^\star) \mid z^\star)] = 0$, because the second factor is $\sigma(z^\star)$-measurable. **That measurability is the condition, and it fails if any argument of g is left out of the conditioning variable**: with $g = g(z^\star, T)$ and T excluded, the cross term need not vanish and the binning in (c) is no longer a coarsening of the conditioning σ-algebra, so neither the identity nor the upper bound survives. Where g reads a temperature—the broad IDAC set of §3.5.1, dim $=$ 103—we accordingly take $z^\star = (\text{profile pair}, T)$; on the VT-2005-matched set every row sits at 298 K and the two readings coincide. The same rule fixes $z^\star$ in the second domain: the Hammett closure of §3.6 reads a per-series constant pair as well as the substituent constant, so there $z^\star = (s, \sigma^\star_{\mathrm{H}})$ and the conditioning is on both. *(b) Jensen lower bound.* With $\Delta(z^\star) = \mathbb{E}(m \mid z^\star) - g(z^\star)$, $B_{\text{closure}} = \mathbb{E}[\Delta^2] \geq (\mathbb{E}[\Delta])^2 = (\mathbb{E}[m]-\mathbb{E}[g])^2$, using $\mathbb{E}[\mathbb{E}(m \mid z^\star)] = \mathbb{E}[m]$. It needs *no* smoothness assumption; its one requirement is that m and g share the $\ln\gamma_2^\infty$ reference state, which the closure validation of §2.3 establishes. *(c) Law-of-total-variance upper bound.* For any binning $B = \text{bin}(g(z^\star))$ one has $\sigma(B) \subseteq \sigma(z^\star)$, and conditioning on a coarser σ-algebra cannot reduce expected conditional variance: $\mathbb{E}[\text{Var}(m \mid B)] \geq \mathbb{E}[\text{Var}(m \mid z^\star)] = B_{\text{insuff}}$. *(d) The estimand is convention-free; the bound is not.* $B_{\text{insuff}} = \mathbb{E}[\text{Var}(m \mid z^\star)]$ is a functional of the law of $(m, z^\star)$ alone and does not involve g, so it is identical across closure conventions and any $\text{MSE}(g_1) - \text{MSE}(g_2)$ is entirely $B_{\text{closure}}(g_1) - B_{\text{closure}}(g_2)$. The bound of (c) is not convention-independent, and no contradiction arises: two conventions coarsen the same $\sigma(z^\star)$ along two different statistics, giving two valid upper bounds on one quantity, of which the minimum is again valid but is biased low in finite samples, so we report it and do not headline it. □

B_{closure} is the systematic discrepancy of a fixed physical map in the sense of Kennedy–O'Hagan,[19,20] and its dominance over B_{insuff} is the empirical content of the claim that the ceiling is set by the closure rather than by the inputs. **What B_{closure} is the error of is a composite object.** The identity compares $g(z^\star)$ against $\mathbb{E}[m \mid z^\star]$, and $z^\star$ is not measured but produced by a procedure—for σ-profiles a quantum-chemical calculation on one conformer and one protonation state per molecule at one level of theory—so B_{closure} is a joint property of the map and of the procedure that produced the reference it reads, and not of the kernel's functional form alone: a displaced reference loads onto B_{closure} through the curvature of g even where g is exactly specified (§S9). The deployed pipeline consumes that same composite, so it is the object a practitioner faces, but every attribution below is to the pair and not to the kernel. Where model-discrepancy calibration *fits* a correction from held-out observations, here g is fixed and $z^\star$ supplied from outside the fit, so the composite's error is *attributed* rather than corrected—available only where an external latent oracle exists. **Exactness is contingent on g being fixed**: for a *fitted* decoder the cross term does not vanish and a plug-in point split is invalid. Because B_{insuff} is bounded above and B_{closure} below, a positive *separation margin* $\text{MSE} - 2B^{\text{up}}_{\text{insuff}}$ establishes that the closure's own error exceeds what input insufficiency can account for, while a non-positive one is a failure to separate and licenses nothing: a reversal would need a *lower* bound on B_{insuff}, which this instrument does not supply. **Its size is a lower bound and not an estimate.** Since $\text{MSE} = B_{\text{insuff}} + B_{\text{closure}}$ exactly, $B_{\text{closure}} - B_{\text{insuff}} = \text{MSE} - 2B_{\text{insuff}}$, and the reported margin under-states that difference by $2(B^{\text{up}}_{\text{insuff}} - B_{\text{insuff}}) \geq 0$, an amount this design cannot measure. Every margin below therefore says the closure's error exceeds its inputs' by *at least* the number printed, and a bootstrap interval on a margin is an interval on that bound: its upper endpoint limits nothing.

What population, and what sets the resolution. The population is the *measurement superpopulation*: a draw is a nominal compound pair *at a temperature*, together with the conformer, tautomer and protonation state actually populated there, and an independent determination of $\ln \gamma_2^\infty$ for it. Neither the conformational state nor the replicate is part of the conditioning variable, so $\mathrm{Var}(m \mid z^\star)$ is the spread of $\ln \gamma_2^\infty$ across the conformational ensemble and across replicate determinations at one reference profile and temperature—nonzero for physical reasons unrelated to sample size. Experimental label noise is a *component* of it and never a term added beside it, and what the design cannot do is *observe* it, neither set holding two rows at a common argument. The bounds are therefore conservative in one direction only—were the superpopulation variance negligible the finding would be the stronger $B_{\text{closure}} = \text{MSE}$—so nothing reported can be an artifact of over-estimating B_{insuff}, no entry anywhere is the share of the error the inputs contribute, and a low cluster-bootstrap frequency P is loss of resolution and never evidence for the inputs. Resolution is governed by $\dim(z^\star)/n$.

2.1.2 The decomposition and the grounding gap

The decomposition is the measurement the conclusions rest on: assumption-light, needing neither a representable encoder nor a lossless bottleneck. Alongside it we report a second, purely observable quantity, the *grounding gap* $\mathcal{G} := R_{\text{orc}} - R_{\text{e2e}}$, the oracle risk minus the best end-to-end risk; the paradox is $\mathcal{G} > 0$. Under a representable encoder (A1) and a lossless bottleneck (A2) it is sandwiched $0 \le \mathcal{G} \le B_{\text{closure}}$, so $\mathcal{G} > 0$ would certify $B_{\text{closure}} > 0$ (Theorem S1, §S2). A1 is unverified—the linear probe of §S6.1 does not bear on it, scoring an untrained encoder of the same architecture within its own spread—and A2 is unverifiable and physically dubious; when A2 fails, $\mathcal{G} > 0$ conflates closure misspecification with input insufficiency, a well-specified closure with a lossy latent giving $B_{\text{closure}} = 0$ yet $\mathcal{G} > 0$. We therefore read $\mathcal{G}$ as corroborating evidence and base the model-dominance conclusion on the decomposition.

The sign rule, and scope. Whether the *trained* physics model outperforms the *trained* black box at sample size n follows a sign rule. Writing each trained risk as an asymptote plus estimation variance, the *physics penalty at budget* n is $\mathcal{T}(n) \approx \Delta_\infty - D(n)$, where $\Delta_\infty := R_{\text{e2e}} - R_{\text{direct},\infty}$ is its *asymptotic* part and $D := V_{\text{direct}} - V_{\text{phys}}$ the variance a constrained prior saves. Physics helps exactly when that saving exceeds the asymptotic bias; and if $\Delta_\infty > 0$ while the variance gap vanishes, there is a critical budget beyond which it does not (Proposition S1, §S2, with the limiting refinement in which $\Delta_\infty = B_{\text{closure}} - \mathcal{G}$ when the control is asymptotically Bayes). The consequence used below is that a prior is predicted to hurt precisely in the mature-data, low-noise, well-covered regime. Each physics-versus-control difference reported here is a penalty at the budget actually trained, not an asymptotic one; there is no "optimal grounding" theorem, and the decomposition does not improve out-of-distribution accuracy. Its variance term $D(n)$ is what carries the semiparametric phenomenon in which an estimated propensity outperforms the known-true one:[21,22] a fitted nuisance can be the more *efficient* estimator even when its bias is no smaller, so a true-valued input need not give the lower total risk at finite n.

2.2 Data, splits, and labels

Solubility measurements come from BigSolDB 2.0:[23] 127,930 rows over 15,665 solutes and 221 solvents between 243 and 438 K, partitioned by Bemis–Murcko scaffold into 111,724 training, 8,103 validation and 8,103 test rows so that every test solute is unseen at training; a melting-point-independent miscibility/structure filter sets row membership, and crystal T_m, ΔH_{fus}, Hansen parameters and infinite-dilution activity coefficients $\ln \gamma_2^\infty$ are merged on as per-solute labels where available. A split's rows do not all carry the same label: of the 8,103 test rows, 5608 carry a solubility measurement and the other 2495 carry a melting point and no solubility, so every accuracy number below is on 5608 rows and says so (§2.4). Crystal supervision is scarce—the

Table 1: Terminology and notation. Each term is defined here and used below without re-glossing.

Term	Meaning in this manuscript
closure g	the fixed, non-fitted physical map from the intermediate to the observable: here COSMO-SAC, in §3.6 a Hammett relation
kernel	the segment-interaction energy inside COSMO-SAC; a closure variant is named for its kernel's year (*2002*, *2010*)
arm	one trained configuration, or one closure variant evaluated; *DirectGNN* is the closure-free control arm and *TGNNSolv* the closure-carrying one, abbreviated *TGNN* in Table 3
$z^\star$, $\hat{z}$	the closure's *entire* argument at its reference (tabulated) value, and the network's own prediction of it; a σ*-oracle* feeds $z^\star$ where $\hat{z}$ would go. In §3.6 $z^\star$ is a pair and only one of its coordinates is learned
grounding	two opposite operations on one reference database: *supervising* $\hat{z}$ against it during training, and *substituting* $z^\star$ for $\hat{z}$ at evaluation
B_{closure}, B_{insuff}	the error of the closure *as deployed*—the fixed map read at the tabulated intermediate, so a property of the map and of the procedure that produced that intermediate *together*, and not of the kernel alone (§3.5.3)—and what its inputs cannot resolve, Eq. (2); B_{closure} is bounded below and B_{insuff} above, so the comparison is one-sided
margin	$\mathrm{MSE} - 2B^{\mathrm{up}}_{\mathrm{insuff}}$, the MSE of Eq. (2)—the closure read at $z^\star$ against its own target m, so in squared m units, squared $\ln\gamma_2^\infty$ on the activity axis and not the solubility axis; positive establishes $B_{\mathrm{closure}} > B_{\mathrm{insuff}}$ and its size is a lower bound on $B_{\mathrm{closure}} - B_{\mathrm{insuff}}$ and not an estimate of it; non-positive is failure to separate and licenses nothing
$\mathcal{G}$	the grounding gap $R_{\mathrm{orc}} - R_{\mathrm{e2e}}$; distinct from Γ_S, the segment activity coefficient inside COSMO-SAC
stratum, map	a chemically defined row set of the broad IDAC set, and the 59 of them the estimator is run on
boundable	$n \geq 40$, so the fixed cell's eight bins hold five rows each
admissible	boundable *and* sign-stable under deletion of each contributing publication, in all four unit $\times$ convention cells (§2.5)
full, deployed	the combinatorial convention: *full* carries the Staverman–Guggenheim term, *deployed* is residual-only and is what this pipeline runs
UD	the University of Delaware σ-profile database
labelled test rows	the 5608 of the 8103 scaffold-split test rows carrying a solubility measurement; the other 2495 carry a melting point only and are scorable in no arm
broad IDAC set	477 IDAC $\cap$ UD rows, 185 pairs, all temperatures, UD profiles
VT-2005-matched set	60 pairs at 298 K carrying a VT-2005 profile on both sides
σ	alone, the screening-charge-density profile; $\sigma(\cdot)$ with an argument, the generated σ-algebra; σ_{H}, the Hammett substituent constant, $\sigma^\star_{\mathrm{H}}$ its tabulated value under the rule of §3.6 and $\hat{\sigma}_{\mathrm{H}}$ the network's; σ_η, experimental label noise

test and validation splits contain essentially no measured ΔH_{fus}, the quantitative basis of the identifiability argument (§2.1). The external single-component data used for grounding are an open crystal-property pool (∼15,000 melting points), the VT-2005 σ-profile database, and ThermoML infinite-dilution activity coefficients.[24] Two hazards the pipeline handles explicitly—melting-point unit detection, and the instability of the seeded scaffold split across pipeline versions—are recorded in §S9.

The label-noise floor, and the two functionals it is read with. Independent determinations of the same system disagree, and that disagreement floors what any predictor can achieve. On the raw BigSolDB rows, group by (solute, solvent, temperature to 1 K) and keep the groups reporting at least two distinct sources; within a group the disagreement statistic is the mean absolute deviation of $\ln x_2$ from the group mean. Aggregating that per-group statistic over groups requires a choice of functional, and the two standard ones differ by a factor of five, so both are reported: the mean over groups is 0.15 $\ln x_2$ units for the 3948 groups backed by two sources and 0.31 for the 349 backed by three, while the median over the same groups is 0.03 and 0.09. Collapsing exact-duplicate values first—the same measurement re-tabulated across compilations, which would otherwise deflate the floor—leaves the mean at 0.16 over 4261 groups. The scaffold-split MAE reported here (1.70 to 1.85) is above the larger functional by a factor of five, so label noise does not account for the accuracy ceiling under either reading. This floor is a mean absolute deviation in $\ln x_2$; the aleatoric estimate published for this task, an inter-laboratory RMSE of 0.75 in $\log_{10} S$ on 34 duplicate solutions,[1] is a different functional on a different base, and the two are neither pooled nor converted into one another anywhere below.

Disclosure: the scaffold guarantee is not certified for these runs. The released stream builder excludes any pool molecule whose scaffold appears in validation or test and aborts on a missing split file—but that is a property of the builder, not a certified property of the runs reported here. One σ-stream build on disk was generated against the wrong split files. Counted inside the stream, 91 of its 1425 rows are held-out solutes by canonical SMILES; counted where such a row can act, 7 of the 147 solutes spanned by the 5608 labelled test rows have their own reference profile in that stream and 8 have a scaffold relative in it, on 297 and 351 of those rows. Solute coverage, not stream-row fraction, is the load-bearing figure for a scaffold split. Per-run stream provenance was not logged, so the artifacts cannot certify which build fed which arm. Each fact that bounds a leak bounds one contrast: the paradox contrast is one checkpoint evaluated two ways, and DirectGNN uses no stream. The grounded-versus-ungrounded supervision contrast is bounded by neither, because the ungrounded arm carries no σ stream either, so the only arm a leak can reach there is the grounded one, and the 0.20 gain of §3.1 stays uncertified until the leak-free re-run. §S9 gives the file evidence, each fact with the contrast it bounds, and the one molecule the leak touches in the n=44 analysis of §3.3. We therefore do not assert the scaffold guarantee for these runs.

What the leak-free re-run reports, and what it decides. The re-run is specified before its numbers exist, and its protocol, its decision rule and the list of displays it rewrites are set out in full in §S9. It trains the ungrounded and the grounded arm at five seeds (42 through 46) under the published COSMO-SAC configuration, unchanged, on a σ stream rebuilt against *this* split's files and certified inside each training run—the run re-reads the stream and the split files and asserts that they share no molecule by canonical SMILES and none by Bemis–Murcko scaffold, and writes both SHA-256s into its manifest—and reads accuracy on the same 5608 labelled test rows under the same cross-arm intersection rule (§2.4), reporting five per-seed values per arm rather than a mean alone. The decision is fixed in one direction only, since the supervision contrast alone hangs on it, and in three branches: if the five per-seed gains are of one sign the two senses of grounding keep their opposite signs and the published values are replaced by the re-run's; if they straddle zero, supervision has no established sign at five seeds and the abstract, §3.1, the Conclusions and Table S2 are rewritten to report one operation with a sign and the other as unresolved; if they are of one sign and it is the opposite one, that is what is reported. Which displays the five-seed values replace is fixed here too rather than chosen afterwards—the three arms that carry those values, every float and passage resting on them, and the three numbers that instead keep their published value with the five-seed one beside them because the re-run supplies one side of the comparison and not the other—together with the boundary in both directions: neither the DirectGNN control nor the NRTL arm is retrained, and §3.5 uses no trained model at all. The substitution result is unaffected in all three branches: it is one checkpoint evaluated two ways, and no σ-stream build can reach it.

2.3 Model, closure, and training

A *shared* message-passing encoder embeds solute and solvent, a bipartite cross-attention block lets them interact, and the resulting pair representation feeds every head. Two further backbones are implemented and neither carries a number reported here; the physics-channelled one was run against this encoder and did not beat it, an observation recorded with no set, seed count or interval attached to it (§S1), so no accuracy comparison between backbones is claimed. A fusion head predicts T_m and ΔH_{fus}, giving the ideal term $\Phi(T)$ of Eq. (1). The activity term $\ln \gamma_2$ comes from one of three differentiable closures. Two are controls, the fitted NRTL model and its symmetric one-parameter collapse. The third, parameter-free, is the closure under study: a σ-profile head emitting a 51-bin screening-charge-density profile per molecule, and a COSMO-SAC layer mapping the profile pair to $\ln \gamma_2$ with an optional Staverman–Guggenheim combinatorial term—the *full* versus *deployed residual-only* distinction on which §2.5 and §3.5 turn. The saturated mole fraction solves $\ln x_2 = -\Phi(T) - \ln \gamma_2(x_2, T)$ as a damped fixed point in x_2; a bounded, gated adaptive correction perturbs the physical parameters rather than the output, so the physical decomposition survives it.

External single-component data enter through *grounding streams*—the training-time sense of grounding. Each stream is a sidecar of *self-solvent* rows (no solubility label, only the target factor's mask active) interleaved into training under the guard above, grounding the crystal factor from the melting-point pool and the activity factor from the σ-profile database, and so identifying from external data a factor that solubility alone leaves non-identifiable and that a black-box predictor cannot use. Training runs a three-stage curriculum under a weighted loss: a warm-up on the auxiliary streams alone, then joint training against solubility, then a low-rate stabilisation stage. The schedule pinned for the arms of Fig. 2 freezes the σ head at the end of the warm-up, so the σ stream enters the warm-up only.

The in-house closures, and where their validation stops. The dispersion-off closures in the fidelity measurement (§3.5.3) are our own differentiable implementations, so those arms share one code path, and each consumes its own model-prescribed σ-averaging (Mullins for 2002, Hsieh for 2010). Our 2002 layer—the *fair 2002* arm wherever the kernels are compared, because it reads all three profile channels—reproduces the NIST reference COSMO-SAC-2002[25] to RMSE 0.0035 $\ln \gamma$ on identical VT-2005 profiles, and our COSMO-SAC-2010/dsp layer,[26,27] built on the *typed* latent of three donor/acceptor-resolved σ-profiles per molecule in place of one, reproduces the NIST 2010 reference with dispersion *off* to RMSE 1.4×10^{-4} $\ln \gamma$—that second figure quoted from its validator's console, which wrote no artifact where the 2002 one deposits its comparison, so it cannot be checked against the deposited record (§S1). The dispersion term is *not* validated to that standard, which is why every dispersion-off contrast is computed on our layers while the *2010, dispersion on* runs are quoted from `cCOSMO`, that reference implementation itself.

The same closure against a published evaluation. Reproducing a reference implementation fixes the code; it does not say whether the error this paper decomposes is the error a published COSMO-SAC has, or something peculiar to this data assembly. Table 2 answers that. De Souza et al.[28] score the original Lin–Sandler COSMO-SAC on the infinite-dilution activity-coefficient database distributed with *The Properties of Gases and Liquids*, at AAD $= \mathbb{E}|\ln \gamma^{\infty}_{\mathrm{calc}} - \ln \gamma^{\infty}_{\mathrm{exp}}|$ of 1.75 with HF-TZVP profiles and 0.68 with BP-TZVPD-FINE: one closure, two profile databases, a factor of 2.6 between them. An evaluation of COSMO-SAC is in that sense an evaluation of the profiles it was handed, which is why B_{closure} is defined above on the composite of map and profile calculation and not on the kernel. We ran our own differentiable layer over the same records (§S9), and on AAD it lands inside the bracket the published closure spans across profile databases, under both conventions; on R^2 it is inside in the deployed convention and level with the bracket's lower end in the full one, the two printing alike at 0.88. This paper's own two sets are inside on AAD alone, and in the deployed convention only—their R^2, 0.61 and 0.40, is below the published range. Two limits stay open: the profile database is the one axis we cannot match, VT-2005 being what this paper deploys and the axis that publication shows dominates; and the publication's 6977 points are not proven to be the same rows as the 7889 scored here, the record selection not being stated file by file. Subject to those, the closure error decomposed in §3.5 is, in AAD, the error a published COSMO-SAC has; which half of this paper's chemical localisation carries across to that set, on 130 times the data and someone else's compilation, and which half does not, is in §3.5.3.

2.4 Evaluation protocol, and the DirectGNN control

The *oracle substitution* replaces the predicted σ-profile $\hat{z}$ with the reference VT-2005 profile $z^\star$ before the closure, matched by canonical SMILES. What is replaced is one object and not two: the σ-head emits a normalised 51-bin shape and a cavity area whose product is the area-weighted profile the closure reads, and for a matched molecule that pair is replaced by the tabulated pair, which stands in the same relation—so the reference area reaches the residual term's A_2/a_{eff} prefactor as part of replacing that one object rather than as a separate choice. Solute and solvent are matched independently. Nothing else is replaced, the crystal head reading a solute-only embedding the substitution does not reach; the bounded correction does read the closure's own parameters, so the crystal values the solver finally receives shift by a little, at most 7×10^{-4} in Φ against a median magnitude of 4.7. §S1 lists every replaced and unreplaced quantity, and gives what the match rule reaches: the reference table covers 99.3% of the labelled test rows on the solvent side and 5.3% on the solute side, so the substitution contrast below is carried by the solvent profile.

It is applied three ways to rule out implementation artifacts: at evaluation only; injected at training time so the model co-adapts; and as a channel-swap control, injected at training under a coordinate-descent schedule that freezes the crystal branch during the SLE phase, so that only the activity branch refits against it. The three differ in when the replacement is made, not in what is replaced. The last two are at seed 42 only. An analogous override substitutes reference crystal targets. The activity target for the decomposition is the infinite-dilution $\ln \gamma_2^\infty = \lim_{x_2 \to 0} \ln \gamma_2$, on two sets that differ in their reference database and their regime (§3.5.1–§3.5.2).

DirectGNN shares the encoder, interaction, readout and pair representation, adds a thermometer (cumulative-threshold) temperature encoding, and maps directly to $\ln x_2$ with no heads, closure or solver. The two arms therefore share their representation but not their inputs or their tuning: the control receives an explicit temperature encoding the physics arm does not, seeing T instead through $\Phi(T)$ and the closure, and the optimisation settings were tuned per arm and differ by up to an order of magnitude. The TGNNSolv$-$DirectGNN gap is accordingly a comparison of two separately tuned pipelines that differ in the thermodynamic bottleneck, not an isolation of the bottleneck, and every physics-penalty comparison below carries that caveat (§3.7).

The controlled comparisons of §3.1 are run at three seeds and reported as mean±sd on a fixed row set; 5608 is that lock—the test rows that carry a solubility measurement, 5608 of the 8103 (§2.2)—so all arms score identical rows. The 2495 rows outside it carry a melting point and no solubility and are unscorable in every arm alike, the control included; none is dropped for anything an arm predicted. The set is fixed by which rows carry a label and not by which rows an arm handled—the lock's finite-prediction clause is inert, removing no row from any arm at any seed—and §S3.2 gives the decomposition. The runs behind every number reported here fall into twelve families, a family being a set of runs sharing a script, a corpus and a seed policy; Table S25 gives each number its family, its seed count and the artifact it regenerates from.

Table 2: The deployed closure against a published evaluation of the same closure. AAD is the mean absolute error in $\ln\gamma^{\infty}$. The middle block's per-cell and per-regime values are in §S9.

COSMO-SAC-2002 on	profiles	n	AAD	R^2
As published[28]				
PGL 6th ed. IDAC set	HF-TZVP	6977	1.75	0.88
PGL 6th ed. IDAC set	BP-TZVPD-FINE	6977	0.68	0.96
The same set, through this paper's closure				
full convention[a]	VT-2005	7889	0.98	0.88
residual (deployed)	VT-2005	7889	0.77	0.92
non-aqueous rows	VT-2005	4347	0.47	0.68
aqueous rows	VT-2005	3542	1.13	0.83
This paper's own sets, residual convention				
broad IDAC set	UD	477	1.12	0.61
VT-2005-matched set	VT-2005	60	0.81	0.40

[a] The like-for-like row, a published COSMO-SAC carrying its Staverman–Guggenheim term; the residual row is the convention this paper's B_{closure} is measured in.

2.5 Estimating the split: one cell, one convention, one rule

Four analyst choices set the law-of-total-variance bound—the bin count, where the equal-count bin edges fall, the within-bin variance convention, and the combinatorial convention—and all four are fixed here, before any margin is reported.

The estimator cell. Eight equal-count bins of $g(z^{\star})$, the *unbiased* within-bin variance, and the row unit, applied identically to every set and every stratum whatever its size. The bin count is the choice with the most leverage—it takes the broad set's aggregate margin from $+0.01$ to $+0.96$—so the cell is not left standing alone anywhere a finding rests on it: the same sweep is run for the strata the admissibility rule passes, and reported with them. Enumerating every equal-count contiguous partition, the bin-edge choice is immaterial on the broad IDAC set and material on the VT-2005-matched set, whose margin changes sign inside the envelope; we accordingly read that set's margin as consistent with zero and carry the claim on the broad set in stratified form.

The convention rule. The margin depends on the combinatorial convention, so one rule governs both sets: headline the *deployed* residual-only convention, pairing MSE and the bound computed under that same convention. Headlining the deployed convention on the smaller set and the full convention on the broad one would be the favourable choice on each set taken separately; one rule applied to both costs the broad set's margin about half its value. The Jensen constant-offset bound is convention-specific—it separates under the full convention and inverts under the deployed default—so what is reported rests on the model-free law-of-total-variance bound, which fits nothing and so lets no held-out row borrow strength from a near-twin. Both bounds are valid on the one population quantity, by Lemma 1(d), and so is the smaller.

The stratification rule. The broad set is stratifiable where the VT-2005-matched set's 60 pairs are not, on two axes that are pure functions of molecular structure: an ordered nine-class cascade of substructure tests over the solvent, hydrogen-bonding role first and dominant functional group second, and the solute's role as water or as an organic. The classifier reads a SMILES string and nothing else—never m, g, the squared error, the source or the bootstrap—so no cell can have been chosen on its outcome, which is a property of the code rather than an assurance. A stratum is *boundable* at $n \geq 40$, where eight bins hold five rows; the smaller ones are reported at that same cell, with the adaptive bin count $\max(3, \lfloor n/10 \rfloor)$ in the rows n it is applied to carried beside them in the deposited table, nothing being dropped for being small.

The admissibility rule. A stratum is *admissible* only if it is *(a)* boundable at that fixed cell and *(b)* robust to leave-one-source-out—its margin keeps its sign when each source publication contributing to it is deleted in turn—in all four unit $\times$ convention cells, since the map reports all four. That last clause is load-bearing and is never dropped: without it the set taken whole passes too. A stratum drawing on one publication fails (b) by non-testability, there being no deletion to perform; a deletion leaving fewer than 16 rows leaves the fixed cell undefined, and fails (b) as well. A third clause reduces the admissible *cells* to row sets, and it is stated here with the same precision rather than left to prose, because the one claim this paper carries from the map and the multiplicity null that tests that claim both turn on it. *(c)* Admissible cells holding identical rows count once; and an admissible positive row set X is *demoted* to another, Y, when Y is admissible and positive, Y holds more rows than X, more than half of X's rows are Y's, and the residual $X \setminus Y$ is not itself admissible—so that X's margin does not stand outside Y. What decides it is that residual and not the overlap: a row set most of whose rows lie inside a larger one survives (c) whenever its own margin stands outside that larger one, so *(c)* is not containment and no cell is demoted for overlapping (§S3.3). Clause (c) takes the six admissible cells of §3.5.1 to three distinct row sets; positivity then removes one of the three and (c)'s residual test the second, leaving one. It is rebuilt on every draw of the multiplicity null, on the permuted labels, like the rest of the rule (§S3). Everything else is *reported in the map and not stated*, with its n, its sources, its margin, its interval and the reason. The rule governs the cells that favour us and those that do not alike, and it tests whether a number is stable, not what it means: an admissible stratum whose margin is not positive, and one that (c) demotes, both establish nothing.

3 Results and discussion

A cluster-bootstrap frequency P is the frequency, over two-way (solute $\times$ solvent) cluster resamples, with which the margin it is attached to stays positive; it is not a p-value against any null. Such frequencies are quoted as bands—rounded to one decimal, with > 0.95 and < 0.2 closing the ends—because with 17 solvent and 41 solute clusters on the VT-2005-matched set they do not carry two-decimal resolution; the two-decimal values are in the deposited artifacts. The resampling intervals of this paper—the Abstract's, the ranking and recalibration intervals of §3.2, the margins of §3.5 and the pKa intervals of §3.6—are *percentile* intervals: their endpoints are order statistics of a resampling distribution, with no bias correction and no studentisation, over the unit named beside each. §3.2 resamples the 107 solute clusters, 4000 draws, at the 95% level; §3.5 resamples solute and solvent clusters two-way, 3000 draws; in §3.6 the decomposition bounds resample molecules within each Hammett reaction series and the flip gap resamples the four series themselves, 4000 draws. Not every $\pm$ or bracketed span below is one of those. Two in particular are not: the kernel-transfer figures of §3.5.3 are a standard *error* over five fit seeds, and the leave-one-publication-out span of §3.5.1 is the range of that curve. Every claim below carries its set, its size and its scope in the sentence that makes it; a reader who wants all of them side by side will find the same eighteen collected as one table in §S3.1 (Table S2).

3.1 Supervising the intermediate toward the reference, and substituting it at prediction time

The word "grounding" names two opposite operations on the same VT-2005 database. Every solubility figure in this subsection and the next is an $\ln x_2$ MAE over one row set, the 5608 of the 8103 scaffold-split test rows that carry a solubility measurement (Table 1); the single-seed control arms below carry that identical set, so nothing here is compared across row sets. Supervising the σ head against the database *during training* improves solubility MAE from 2.043 ± 0.040 (ungrounded) to 1.846 ± 0.053

(grounded), a gain of 0.20 over three seeds, the two arms' per-seed values not overlapping. The ungrounded arm carries no σ stream, so this is the one contrast in the paper that the uncertified stream build of §2.2 could inflate rather than one the arguments there bound (§S9). Substituting the reference profile for the learned one *at inference* degrades it, and that substitution alone is the effect this paper is named for. It erases any positive R^2 (Fig. 2): the model predicts solubility better *from its own learned σ-profile* than when *given the external reference one* (MAE 1.85 vs 2.25, in $\ln x_2$ units, over the same three seeds and with no overlap between the two arms' per-seed values either). Of the figure's eight arms two are non-COSMO controls, DirectGNN and a fitted NRTL closure; five COSMO-SAC arms differ only in how the σ head is trained and in what it is fed, and the sixth (*grounded+comb.*) differs instead in the closure, restoring the Staverman–Guggenheim combinatorial term. *Ref. σ (eval)* is the grounded checkpoint re-evaluated with the reference profile substituted at test time, *ref. σ (train)* the co-adaptation control injecting it during training instead.

The headline +0.41 is the evaluation-only substitution, and part of it may be unrelated to the closure: swapping an input distribution at test time degrades a pipeline on its own. The control that removes that component injects the reference profile *at training*, so the model co-adapts; at seed 42 it costs +0.18 ($1.803 \to 1.981$) where the evaluation-only substitution costs +0.43 at that same seed ($1.803 \to 2.232$), so part of the headline gap goes with co-adaptation and the co-adapted arm is by design free of *that* confound. That part is a share of 0.59 at seed 42, and we report it as a description of that seed rather than as a magnitude: its numerator was measured once, and reading the same numerator against the other two seeds' baselines gives 0.83 and 0.61 (§S3). A channel-swap control costs +0.27, to 2.08. It carries a second confound the co-adapted arm carries too: neither is certified to have trained on the same σ-stream build as the seed-42 comparator both are read against, and the file times settle nothing either way (§S9). Both are worse than the learned-σ arm, which rules out an implementation artifact. Only the three-seed arms can be set against one another, and there the evaluation-only substitution costs about twice what supervision gains (+0.41 against −0.20). That ratio is between two three-seed arms and carries the co-adaptation confound with it, so it is not the paradox's co-adaptation-free magnitude; the arm that is, at +0.18, was run at one seed on an uncertified stream build and cannot be ordered against the −0.20 at all, whose own per-seed values (0.237, 0.075, 0.281) straddle it. Which of the two operations is larger is therefore unresolved, and only their signs are carried forward.

3.2 What the substitution costs: level, spread, and the choice of solvent

Figure 3 sets the same rows out predicted against measured, for the control and three of the closure arms. Two things are visible there that the arm ordering is not. First, every arm is severely shrunk toward the corpus mean: the ordinary least-squares slope of predicted on measured is 0.515 for DirectGNN, 0.416 ungrounded, 0.528 grounded and 0.690 with the reference profile—all four at seed 42, the figure's seed, with no replicate behind them—none near 1, and in every panel the median trace crosses the identity—sparingly soluble compounds are called more soluble than they are and highly soluble ones less. That shrinkage is not the closure's doing: at that seed the closure-free control's slope differs from the grounded closure's by 0.013, so whatever produces it lives in the encoder and the data. Second, the reference-profile arm is the *least* shrunk and the worst. Its slope is the closest of the four to 1 and its prediction spread exceeds the measurement's, yet its MAE is the highest and its R^2 is not distinguishable from zero; substituting the reference restores spread that is not aligned with the truth, and its worst region is the sparingly-soluble tail it appears to reach. That is the misspecified map made visible on the solubility

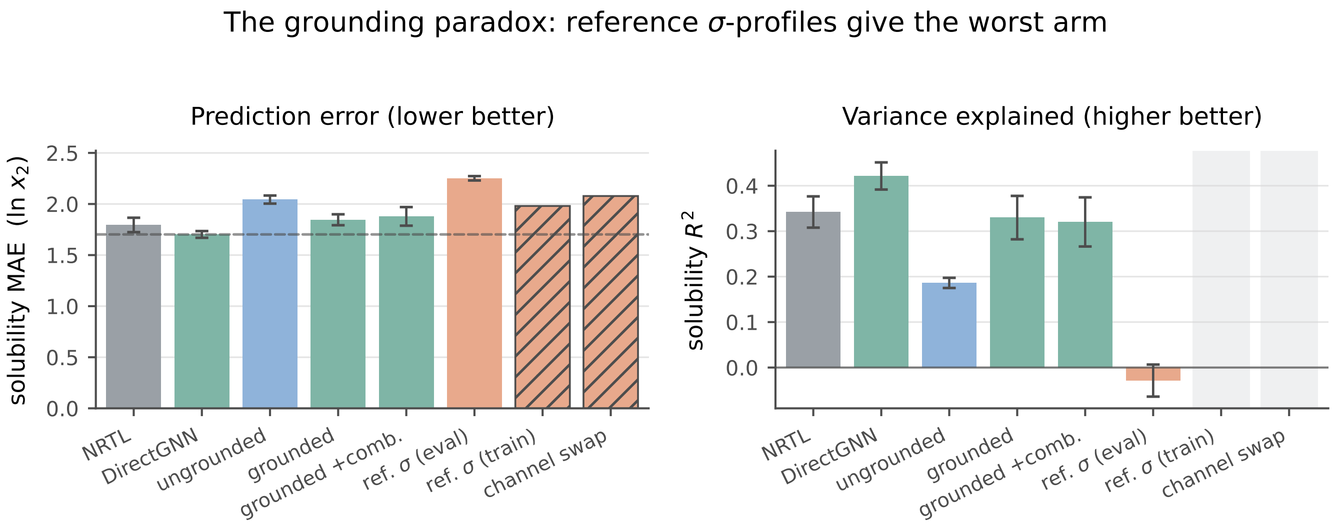

Figure 2: Controlled comparison on the 5608 labelled test rows of the scaffold split. Solid bars: three seeds, mean ± a population standard deviation over them. The two hatched bars are the co-adaptation controls, seed 42 only, so they carry no error bar and no R^2, and keep an empty slot in the right panel. Dashed in the left panel only, the DirectGNN reference.

axis.

The substitution also costs the decision the model is used for, and here the separate-tuning confound of §3.7 does not stand in the way: the reference-profile arm re-runs the grounded checkpoint's own weights over the same rows and changes only the profile fed to the closure, so the two sides differ in Φ by 4.2×10^{-4} at the worst seed and no second tuning stands between them. The confound it does not remove is co-adaptation: like the headline MAE this is an input swapped at evaluation, no arm that had trained on the reference profile was ranked, and what co-adaptation would do to an ordering is unmeasured here. Grouping the labelled test rows by solute and temperature and ranking the solvents inside each group (823 groups over 107 solutes; Table S19), the mean per-solute Spearman correlation falls from 0.592/0.546/0.598 to 0.365/0.369/0.410 across the three seeds—pooled -0.197 $[-0.282, -0.117]$, every per-seed interval excluding zero, and every one of the three larger than any difference between seed replicates of a single arm, the largest of which is 0.060. Every interval here is a 95% percentile bootstrap over those 107 solute clusters, the effective sample rather than the 823 groups. nDCG@3 falls by -0.084 $[-0.124, -0.043]$, and top-1 accuracy by -0.107 $[-0.192, -0.020]$. That last figure is quoted pooled and only pooled: two of its three per-seed intervals span zero, so it is the one of the three ranking metrics that does not hold seed by seed. The solvent the model would select changes in 40 to 53% of groups, asymmetrically: the learned profile alone picks the measured best solvent in 20.0/23.2/21.5% of groups, the reference profile alone in 6.2/14.5/12.2%. None of that is a level. Fitting a separate affine recalibration inside every group—a slope as well as an intercept, the largest concession a reading on which the substitution merely miscalibrates can ask for—removes 86% of the MAE gap, $+0.431$ $[+0.308, +0.553]$ to $+0.060$ $[+0.021, +0.101]$, which is the restored, misaligned spread of the paragraph above in another currency. That map is fitted on the rows it is then scored on, two free parameters against a median of five solvents, so it bounds what the concession can buy rather than measuring it: refitted leaving out the row it scores, it removes none of the gap, $+1.09$ $[+0.3, +2.2]$ pooled, and on the 550 groups holding at least five solvents, where a refit has something to work with, $+0.28$/$+0.07$/$+0.04$ by seed with two of the three intervals spanning zero (§S6.1). The MAE penalty is level and scale in the sense that a per-group map fitted on the answer absorbs most of it; out of sample nothing that sharp can be said in either direction. The ranking penalty is not, under either fit, and by invariance rather than by any centring argument—though the invariance is to an *increasing* affine map, and the fitted map decreases in 10 to 12% of learned-profile groups and 17 to 19% of reference-profile ones, where it reverses that group's order outright. Restricted to the 589 groups of 823 whose map increases in both arms at all three seeds, the ranking gap is ρ -0.148 $[-0.218, -0.079]$ and nDCG@3 -0.068 $[-0.110, -0.026]$, with top-1 then spanning zero.

What stands in its place cannot be told apart from a solute-blind ordering. Set each arm against two references that know nothing about the solute—the same arm's own predictions for a randomly drawn other solute over the same solvents, and a ranking of those solvents by their leave-one-solute-out mean measured solubility—and the learned-profile arm is above both on Spearman, Kendall, top-1 and nDCG@3 at all three seeds, twenty-four intervals of twenty-four excluding zero. The reference-profile arm's own twenty-four margins are small and all but one positive, and it is their intervals that carry the reading: twenty-four of the twenty-four include zero, or twenty-three, one cell sitting on the boundary at $[-0.0008, +0.0762]$ and changing side with the bootstrap draw (Table S19). What that measures is a failure to separate the reference-profile arm from orderings built without the solute, not a demonstration that it cannot be separated from them. Neither arm is chance, the within-group permutation floor being $\rho = 0.00 \pm 0.02$ for both. Supplying the true physical intermediate does not leave the model miscalibrated about solutes; it leaves its ordering of them indistinguishable from one drawn without them.

3.3 What moved inside the model, and whether a downstream correction repairs it

A candidate *mechanism* for the low-fidelity, end-to-end regime is this: a σ-profile head trained *end-to-end* through a fixed, misspecified closure—with no σ supervision in that phase—is under pressure to *cancel* the closure's error rather than to be physically faithful. The hypothesis has two parts and this section establishes only the first. Its antecedent—that the latent leaves the reference along a direction shared across molecules rather than as per-molecule noise—is measured. Its consequent—that the departure cancels part of B_{closure}—is not, and the one direct check available argues against it. The external reference that fixes the ceiling makes the first part testable: we compare the predicted profile $\hat{\sigma}$ against the reference VT-2005 profile molecule by molecule (n=44). That comparison is the anatomy of the object the two operations of §3.1 act on—how far the learned intermediate has travelled from the reference that supervises it and then replaces it—and it is placed here, inside the phenomenon, for that reason and not as evidence for why either operation has the sign it does.

What is measured, and on what. The comparison is within a single grounded model, checkpointed before and after solubility fine-tuning, over the n=44 matched molecules and repeated at three seeds. Those three runs are *not* the arms that produce the paradox: they are a separate set, trained with the σ head left unfrozen through fine-tuning, where the four COSMO-SAC arms of §3.1 hold the head from P2 onward—how far *their* latent drifted is not measured and cannot be recovered from the deposit. What follows is therefore measured on runs of its own and is not established of the checkpoints whose paradox it is offered to explain. Under σ supervision the head reproduces the reference profile to within $36 \pm 1\%$, so the intermediate *can* carry the physical shape—though only as an in-sample fit, all 44 probe molecules lying inside the σ-grounding pool. Solubility fine-tuning then drives it a further $41 \pm 3\%$ away, for a total departure from the reference of $51 \pm 2\%$—a magnitude reported but not stable across analysis configurations, so what these runs fix is the sign and the shared direction below and not the size. A relative deviation carries no scale of its own, and the ladder

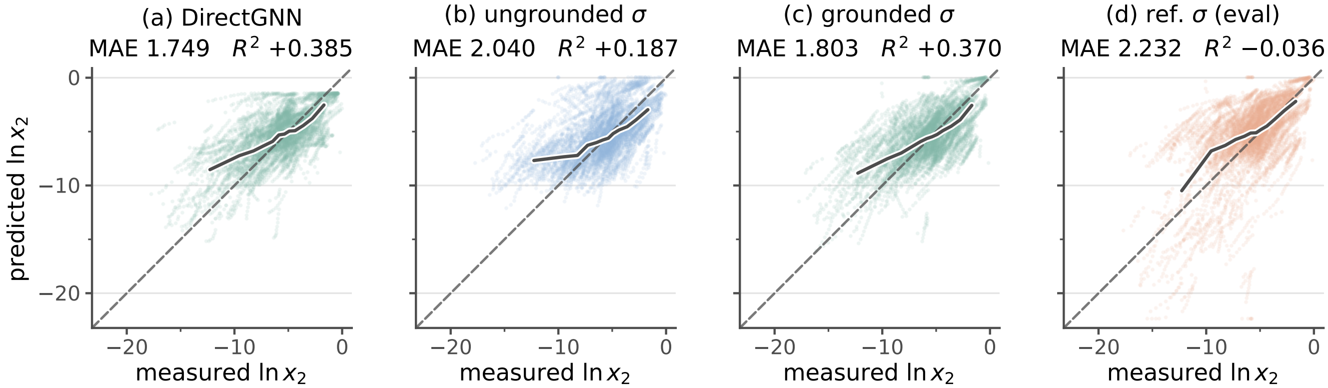

Figure 3: Solubility parity on the scaffold-split test rows: predicted against measured $\ln x_2$ for four arms of the controlled comparison. All four panels are the same 5608 labelled test rows of the scaffold split, from seed 42 of three. Those rows are 763 solute–solvent pairs over 147 solutes and 70 solvents, measured between 248 and 368 K, so one pair contributes a short temperature series and the number of points exceeds the number of independent systems. (a) is the closure-free control; (b) is COSMO-SAC over an unsupervised σ-profile, (c) the same closure with σ-profile supervision during training, and (d) the arm of (c) with the reference VT-2005 profile substituted at evaluation. Dashed, the identity; solid, the median prediction in twelve equal-count bins of the measurement. The MAE and R^2 above each panel are that panel's own, on those rows and that seed, at the three decimals a per-seed value is printed to; they differ from the corresponding three-seed means by at most 0.05 in MAE. Only (d) carries scatter below the identity that reaches $\ln x_2$ values no measurement occupies. The axes span the union of measured and predicted values over all four arms; no point is clipped.

that supplies one (§S5: the same 44 molecules and the same relative-L_2 metric, its rungs built from the reference profiles the grounded arm was supervised on) places the fine-tuned latent at 0.507 and the grounded base at 0.364, against 0.493 for one corpus-mean profile reused for every molecule—a rung carrying no molecular information at all—and 0.651–0.684 for each molecule given another molecule's reference profile. Read on that ladder, the drift consumes essentially all the molecular specificity grounding had provided, which is what makes the shrinkage reading below more than a formality.

What the departure is, and what it is not. The drift is not per-molecule noise and its structure is not a constant offset: the first two principal components of the mean-centred deviation carry 72 ± 1% of the variance, and the quantity read against a null—the top-two share of the learned-versus-reference deviation, 0.399—clears a phase-randomised null (0.221, $p < 0.001$), on one grounded checkpoint rather than over the three seeds, the 0.221 carrying a spread of ±0.01 that is the null's own over draws and not a spread over seeds. Two nulls it does not clear travel with that: the mean-centred share has no matched null of its own, and a wrong-molecule null built from real reference profiles does *not* clear (0.772, $p = 1.00$), so if the case for preferring the phase-randomised null is not accepted the spectrum is uninformative rather than supporting. One distortion operator fitted on half the 44 molecules predicts the drift on the other half 4.1 ± 0.5× better than the zero-drift null—the ± spanning the three seeds at one partition of the 44 and not the partitions—a deliberately weak null, uncontrolled on these three seeds, against which a molecule-independent constant offset already reaches 2.1× on the one companion run where that control can be computed, so the ratio stands against the identity null alone and structure *beyond* an offset rests on the mean-centred spectrum. The held-out half is also homologue-saturated—six n-alkanes, four halobenzenes, five pyridines and four butyl and hexyl amines among the 44, and the median held-out molecule sits 16% of its own profile norm from its nearest training-half neighbour—so "outside the fit" is largely within-cluster interpolation. That spectrum does not exclude a shrinkage of every profile toward the corpus-mean shape, which is how the drift looks on the reference ladder of §S5. The dominant mode (Fig. S4) moves area out of the nonpolar bins and into the polar wings: fine-tuning makes the network represent molecules as more polar than its *own grounded profile* does—the plotted drift being taken against that profile and not the reference, and the mode defined only up to sign, drawn in the mean drift's direction.

Whether the departure cancels closure error. It is not shown to, and the one direct check runs against it. Holding the cavity area at its reference value, the sole retained end-to-end SLE-trained COSMO-SAC checkpoint scores 18.75 against 1.47 on the n=60 VT-2005-matched pairs: on the activity axis its drifted profile is an order of magnitude *worse* through the same closure. Three limits sit on that number—the checkpoint is the uncontrolled one, whose 82% departure is half again the three-seed 51 ± 2%; the axis is wrong, the drift being produced at finite composition and scored at infinite dilution; and the check has almost no power in the direction that matters, so it separates "no cancellation" from "large cancellation" but not from "some cancellation". A second, independent check on the closure's own axis points the same way (§S5). Both sit on that infinite-dilution axis, so neither refutes cancellation in the regime that generates the drift; they remove the one place it could have been measured, and the cancellation reading remains inferred from the paradox itself. What is established is weaker and still consequential: the model's "physical" intermediate is not the molecule's charge distribution, consistent with cautions that the interpretability of physics-grounded latents is not guaranteed under misspecification,[20,29] and that adapting a representation can itself collapse an otherwise physically-faithful latent.[30] Its *structure*—low-rank rather than diffuse—excludes a random collapse but not the structured shrinkage above. The full measurement, its nulls and the two controls are in §S5.

What leaves σ under-determined by activity, and what freezing it buys. The freedom that would make such a cancellation possible is the closure's geometry, measured in §S5: at infinite dilution the activity map concentrates on one or two hydrogen-bonding directions, so the end-to-end head can move σ along the remaining, low-sensitivity ones at little cost to the fit—the σ-profile form of Lemma S1. Nothing there shows recovery of the physical profile to be impossible in principle. The sign of grounding turns on *both* axes, and on which sense of grounding is meant. Freezing the σ head at its supervised warm-up state instead of letting it drift end to end cuts the reference-input penalty by 65% (ΔMAE from +0.40 to +0.14 over n=5571 matched test rows). Those are a separate matched pair of runs—one base resumed twice with the freeze off and on, so a single pair carrying no seed replicate, and the 65% is a difference between two runs and not a mean over any—and not the +0.41 headline arms of §3.1. A near-physical intermediate makes the pipeline far more robust to substituting the reference σ. Supervision alone does not, however, *flip* the sign here—the penalty stays positive, the closure still being the low-fidelity COSMO-SAC-2002 and the warm-up σ itself only partly physical. The full flip needs supervision *and* fidelity together, which is what TeNNet-SAC[7] has and this pipeline does not.

Whether a downstream correction repairs it. If the closure binding were merely an inaccurate point estimate, a learned output-space correction on top of the physics prediction should recover accuracy. It does not: an additive bounded residual with a learned confidence gate leaves the gate effectively shut, and forcing it open removes the systematic *bias* (+0.45 $\rightarrow$ +0.06) but recovers little *accuracy*

(R^2 0.236 $\to$ 0.274 on the same rows), as expected if the closure, not the estimate, is the binding constraint. Both arms are seed 42 only, and this is a statement about the one correction tested: an additive bounded residual on the output, gated, on this closure and this corpus.

3.4 What the substitution does not decide, and what would settle it

Which of the two channels the substitution fails through is not decided here. Drug-like solutes are essentially absent from the VT-2005 set, so the substitution falls almost entirely on the *solvent* channel, and the both-channel substitution the attribution would need covers only ≈7 solutes; enlarging that set would mean computing new reference profiles rather than looking them up.

Nor does anything above decide *why* the substitution hurts. The reading it invites is that handing the closure a faithful input surfaces a misspecified map rather than repairing it, so that the reference profile is the better input only where the map that consumes it is faithful. This paper states that reading and does not assert it, and the reason is stronger than a test that returned nothing: on the solubility axis the two terms of the split do not separate at all, and the solubility axis is where the substitution of §3.1 was measured. The quantities the reading is about—a fixed map's own error, and what its inputs fail to carry—are therefore not observable on the rows §3.1–§3.3 report.

What follows in §3.5 is a second axis and not an explanation of the first. It is a different reference database, a different error functional, no trained model at all, and a row set that shares none of the rows above; the two are measured in different regimes on largely disjoint chemistry, and no measurement in this paper connects them. Until the connecting experiment is run, "the substitution hurts *because* the map is misspecified" is an inference and this paper does not assert it.

Two experiments would settle it, and both are stated as limitations rather than as prospects: the transfer back to the paradox's own regime—finite composition and all γ, where the substitution is measured, rather than infinite dilution, where the split is—which is limitation (i) of §3.7 and the principal open question here; and a typed σ-profile head trained end to end through a modern kernel, which would show whether the substitution's sign survives a closure of different fidelity, limitation (x). Neither has been run.

3.5 A second axis: bounding a deployed COSMO-SAC against its inputs

The estimator is the one cell fixed in advance in §2.5 and used nowhere else in this paper: eight equal-count bins of the conditioning variable, the residual convention, the within-bin Bessel correction, and the admissibility clauses (a)–(c) that decide which strata may be read at all. All four were fixed before any margin was computed; none of §3.1–§3.4 is measured with them.

The decomposition of §2.1.1 separates a fixed closure's error into a part in the map and a part in its inputs. Its output is a map over chemical strata rather than one verdict for a set, for the reason §3.5.1 establishes, and it runs on two data sets that must not be conflated, the *broad IDAC set* and the *VT-2005-matched set* of Table 1.

One reading rule governs every margin below, on either set and under either kernel, and it is stated once here rather than at each of them. The comparison is one-sided in both directions it can be read: a positive margin establishes $B_{\text{closure}} > B_{\text{insuff}}$, while a non-positive one is a failure to separate the two terms and licenses nothing—it is not a measurement that they are equal, and no cell of the map can ever return "better inputs would help here". The size of a margin is a *lower bound* on $B_{\text{closure}} - B_{\text{insuff}}$ and not an estimate of it (Lemma 1, §2.1.1), so an interval quoted on a margin is an interval on that bound and not on the difference itself. And B_{closure} throughout is the error of the fixed map *read at a computed profile*, so what a positive margin establishes is a property of that composite and not of the kernel alone (§3.5.3). The one estimator cell above serves both sets, both kernels and every stratum, so no cell below was fixed after the number it returned was seen.

3.5.1 The broad IDAC set (n=477)

The closure reads a temperature as well as a profile pair and is evaluated at each row's own temperature, so the conditioning variable is its full argument $(z^\star, T)$—the concatenated UD profile pair with T, 103 dimensions against n=477 ($\dim/n \approx 0.22$)—and the target m is the experimental $\ln\gamma_2^\infty$, so floor and closure error are measured on one data set and one profile database. Its chemistry is small: no solute exceeds 12 heavy atoms (median 7), against a median of 14 and a maximum of 122 for the drug-like solids of this paper's own corpus, and only 0.46% of corpus rows carry a solute from this set. *Broad* therefore names the reach of the γ^∞ corpus the kernel is calibrated on and not coverage of the chemistry a solubility model is deployed against: it covers the solvent side of deployment and not the solute side. It is matched to UD by InChIKey with a first-block fallback, looser than the VT-2005 set's rule; a wrong $z^\star$ inflates B_{closure}, so the looser rule runs against the ordering we claim, and the aggregate margin is unchanged on the 465 exactly matched rows.

Every MSE, bound and margin on this axis is in squared $\ln\gamma_2^\infty$ units, and none of them is on the $\ln x_2$ axis the MAEs of §3.1 are measured on. Under the convention rule of §2.5 the fair 2002 arm scores MSE = 1.902 against $\mathrm{Var}(m)$ = 4.85, the LOTV bound gives $B_{\text{insuff}} \lesssim 0.70$ and hence $B_{\text{closure}} \gtrsim 1.21$: a separation margin of +0.51 for the set taken whole, +0.95 under the full convention and +0.27 collapsed to one row per distinct pair, positive in every cell of the bin-count × variance × convention grid, with two of the three bootstrap clusterings extending just below zero.

That aggregate is not the instrument's output. Two operations reported separately—the pair unit, and dropping the two chemistries the error concentrates in, the glycol-ether solvents and the rows carrying water as the solute—compound badly. The chemistry cut alone takes +0.51 to −0.12 on the 258 rows that remain, the pair unit alone gives +0.27, and the two together give −0.38 on 136 pairs; under the full convention the same pair of cuts takes +0.95 to +0.28 to +0.01. Nor is the aggregate carried by the set: deleting one source publication at a time, `10.1021/acs.jced.7b00114`—37 rows, six molecule pairs, 32% of the set's squared error—takes the margin to +0.18, while each of the other fourteen deletions leaves it between +0.42 and +0.87 (under the full convention that curve is flat, [+0.82, +1.07] against +0.95). That single deletion loses two thirds of the margin and its resolution without reversing its sign; the sign reverses only under the compounded cut. Two caveats qualify the corroborating estimators and not the bound—the out-of-fold ones leak under random folds and are uninformative or split under blocked ones, and the tighter binning-free pair bound samples no between-laboratory component and is therefore not promoted—so together they say the binning bound is slack rather than that B_{insuff} is negligible (§S3). What is reported rests on the LOTV bound (§2.5, Table S9).

The map, and how much of it survives the rule. The instrument runs over 59 strata of the broad IDAC set—477 rows, 185 pairs, 78 solutes, 36 solvents, 15 source publications—on the solvent axis at three granularities and the solute axis at two, plus their cross-tab and the set as a whole. Fifteen have $n \geq 40$ and are boundable, the only ones whose margins the rule can act on; the other 44, including the 24 that carry no estimate at all, are printed with the test each fails in §S3.3 (Tables S11–S15 and Fig. S2), every stratum with its error, its bound, its interval and its share of the set's squared error, and nothing dropped for being small—and no stratum but the one named below is stated as a finding anywhere in this paper. Six of the fifteen are *admissible.* Clause (c) of the rule makes those six three distinct row sets, of which one is both positive and not demoted by its residual test: the glycol-ether solvents. **That row set is the one the admissibility rule leaves standing**, at a margin a chemistry-blind relabelling of the same molecules reaches in 10.1% of draws (below): on 182 rows over 43 pairs from three publications carrying 45% of the set's squared error, the closure's own error exceeds what input insufficiency can account for by at least +2.04, that bound carrying a 90% interval of $[+1.26, +2.87]$ at the deployed convention—+2.94 at the pair unit under the full one—positive in all four cells at $P > 0.95$ in each, and under every single-publication deletion, the worst leaving +1.60 when `10.1016/j.jct.2016.10.013`, which carries 53% of that stratum's squared error, is removed. That interval is the two-way cluster bootstrap run to convergence; at the 3000 draws each cell of the map uses, its endpoints carry a Monte-Carlo spread of ±0.02, so no interval in the map is to be read past that digit. It is not an artifact of one solute family: the 74 rows left when its alkane solutes are dropped keep the sign at +1.40 deployed and +1.69 full, under all three deletions. Nor is it a property of the estimator cell. Swept over the bin-count grid Table S5 runs for the aggregate—the analyst choice with the largest leverage here, since that grid takes the aggregate from +0.01 to +0.96—the stratum is positive in all 56 row-unit cells and all 40 the pair unit defines, the deployed unbiased column running +1.06 to +2.19, and the deletion test passes at every one of the 36 cells where it can be run (Table S6). The occupancy rule that fixed the eight-bin cell would ask for 23 bins at n=182, so the cell the map reads is the coarser and the lower of the two. That sweep varies the coarsening's own settings and not the family of estimator: a pre-declared cross-fitted regressor, tested after this analysis was fixed, returns a *looser* bound and moves this stratum to +1.76 $[+1.02, +2.49]$, at the deployed convention and row unit the +2.04 is read at, leaving it the one admissible positive row set but with fewer cells around it, and moves the chemistry-blind null the *wrong* way. That null is re-run on both sides for that comparison, on the 473 rows the cross-fitted estimator scores rather than the 477 the map is built on, so its baseline column is not the 10.1% and 17% quoted below: against a coarsening baseline of 0.094 on those rows the cross-fit gives $p(D) = 0.137$, and the share of certifying draws that reach the observed maximum goes from 16% to 54%. That declaration bound us to headline the cross-fitted estimator UNCONDITIONALLY if it passed a validity gate that reads no margin, and it passed. **We did not follow that rule**: every margin printed in this paper is the coarsening's. The reason and the two things that should be said about the deviation rather than absorbed—that it runs in this paper's favour, and that it is a choice made after seeing a number, which is what a declaration exists to prevent—are in §S3.4, where the poorer map is also deposited cell by cell and the declared rule can be held against this one. What the sweep establishes is therefore robustness within the coarsening, not across estimators.

What the row set does not have is a multiplicity null it clears, and this belongs beside the finding rather than in a limitation. It is by construction the survivor of a 59-fold search. Under 2000 chemically blind relabellings of the same molecules, on all 477 rows and at the coarsening the map deploys—each solvent and solute keeping its rows and taking another's class labels, the stratum family rebuilt and put through the same rule—the map's six admissible strata are above chance ($p = 0.01$) but its two admissible positive row sets are not ($p = 0.286$): such a relabelling certifies at least one row set in 58% of draws. The size is the sharper statistic, and it is not sharp enough: a certified margin of +2.04 or more arises in 10.1% of all draws, which is 17% of the draws that certify anything, against a median of +1.57 among those. The separation is what the instrument licenses on the rows it was computed on; it is not established as a chemistry-specific effect against the search that found it. Each p here is a frequency and not a bound in either direction: the null's own two biases are measured rather than argued and run against each other (§S3).

What settles a selection objection is the same measurement on rows the search never saw, so we ran one. The out-of-sample set is the PGL 6th-edition IDAC database scored by the same deployed closure on VT-2005 profiles (Table 2): another group's compilation, sixteen times the broad set's rows on a different profile database. The row set, the estimator cell, the admissibility rule and what would count as confirmation in each direction were written down before any margin was computed (§S3). **The chemistry could not be tested there**, for a reason about the data rather than about the number: the cell the finding is about—a glycol-ether solvent with an *organic* solute—has no rows there. *Not testable* is the third verdict the pre-declaration lists and counts as non-confirmation, and it follows from the row inventory alone: what the pre-registration delivers here is the absence of confirming data and not a failed replication. The nearest cell that database does carry is water in glycol-ether and polyol solvents, and it is inadmissible under the same rule that governs the map—the rule refuses it out of sample for the reason it refuses water-as-solute in sample—so its margin licenses nothing and the sign is never put at risk. Over that database's ten solvent classes, stated as description and not as a test (Table S10), no class both passes the rule and separates the two terms, and every margin above the one admitted positive class sits on a class the rule refuses. Out of sample the glycol ethers are neither where this closure fails hardest nor a cell the rule can read. The chemical localisation is therefore what the instrument licenses on the rows it was computed on, and the title does not carry it.

It is also the whole of the separation this paper reports, and the statement carries five boundaries that the phrase "COSMO-SAC error exceeds input error" does not: no other stratum of the map returns one (below); neither data set taken whole does (the aggregate is retired above, and the VT-2005-matched set's +0.05 does not stand, §3.5.2); nothing on the solubility axis does, where the two terms do not separate at all and where the substitution of §3.1 was measured; the error that stands above the bound is COSMO-SAC's *as deployed*, the composite of map and computed conformer that the pipeline runs on rather than the kernel (§3.5.3); and the margin is one its own chemistry-blind null returns at the ninetieth percentile, on the one chemistry no out-of-sample rows exist for.

Which chemistry does not separate. Three of the six admissible cells are that same row set at three granularities. The remaining three are two row sets and neither establishes anything: the alkane solutes are admissible and positive and clause (c) demotes them to the glycol ethers, which hold more rows and hold most of theirs, while the acceptor-only aprotics pass the rule and license nothing, the instrument being one-sided. That is the whole of the uniqueness claimed for the glycol ethers: the only row set admissible, positive and left standing by clause (c)—*not* the only admissible stratum and not the only one sign-stable under every deletion. Which cell each of the remaining strata fails on, and by how much—too few rows to bound, one publication so that the deletion cannot be run, or a sign lost under a deletion—is in §S3.3, and none may be used to explain a reversal. The whole-set row passes at the headline cell and fails once the pair unit is required, so the rule does not by itself dispose of the aggregate there; the composition argument above does.

Why the output is a map. $B^{\mathrm{up}}_{\mathrm{insuff}}$ is nearly stratum-independent and MSE is not. Across the three classes the design can bound—glycol ether, water and the acceptor-only aprotics—the bound spans a factor of 2.0 against errors spanning a factor of 25; adding the three that carry an estimate without being boundable gives 6.8 against 51, that 6.8 being set by a mono-alcohol figure the artifact flags as an invalid bound on a stratum already excluded. The two orderings are unrelated—among those six the glycol ethers carry the lowest bound and the second-highest error—and against 2000 random partitions of the broad set's 477 rows into blocks of the observed sizes of its nine solvent classes, at the same cell, the chemical partition's range (largest minus smallest) in MSE exceeds every draw while its range in the bound is unremarkable ($p = 0.26$); their ratio gives $p = 0.02$. Every stratum's figures are in §S3.3. That is the estimator showing through: every stratum is summarised by eight quantile bins of one scalar, so the bound measures the residual spread of m inside such a bin, a property of the binning rather than of the chemistry, while nothing constrains the error to follow it. An aggregate margin is therefore a composition-weighted average of a term free to move 25-fold offset against one that barely moves, and it will report a binding closure for any set whose composition happens to include a chemistry the closure fails on—a statement about the aggregate, not about the bound, which stays valid in each stratum where it is computed.

3.5.2 The VT-2005-matched set (n=60)

This set carries the paper's other reference-matched analyses, and it is far weaker. All 60 rows sit at 298 K, so $z^\star$ is the profile pair alone and $\dim(z^\star)/n \approx 102/60$ leaves the split behind one-sided bounds; it is strongly unbalanced, methanol occupying 25 of the 60 rows; and its 17 solvents fall in six of the nine classes, none reaching the n=40 a bound needs, so it supports no stratified bound and no stratum of it is stated as a finding. Its aggregate margin is +0.05, and it does not stand: it fails under deletion of the two leverage pairs, under dropping methanol, under blocked out-of-fold estimators, and under 11 of the 60 single-pair deletions, where on the broad set no single-pair deletion moves the sign at all (§S3). Zero-mean label noise can understate a separation but not create it (§2.1.1); what a margin here would bound is a label systematic tracking $z^\star$, and that is *not excluded* on the Bessel-corrected binning (§S3). Of the 60 pairs, 57 also appear in the broad IDAC set at 298 K under UD profiles, so the two are not independent samples—this is the sub-design VT-2005, and therefore the paradox's own oracle, is defined on, and the agreement between the two sets' error orderings is a consistency check rather than corroboration.

3.5.3 Whether a better closure moves the ceiling, and where this one fails

If the ceiling is the closure, the direct remedy is a better closure. The standard 2002 → 2010 step delivers one on the VT-2005-matched set and not on the broad IDAC set, and a block split identifies the factor that carries that reversal. The step more than halves the smaller set's error under the full convention ($P > 0.95$), the convention under which the two kernels' published error figures are reported, and cuts it 42% under the deployed one; on the broad IDAC set (477) none of its three cells establishes an improvement, and none measures an equality either—the step scores −0.99 at $P \approx 0.2$ under the deployed convention and −0.07 at $P \approx 0.6$ under the full one, and, with the dispersion term on, +0.06 at $P \approx 0.6$ under the full convention: a point estimate on the improving side, no better resolved than the two that are not. The broad set leaves the sign open in either direction and does not measure it at zero. The reversal is the *chemistry* and neither the profile database nor the temperature range—with UD profiles, one extraction and one convention throughout, the *2010, dispersion off* arm wins on the 57 pairs shared with the smaller set (1.78 → 0.77 at $P > 0.95$, full convention), which are broad-set rows as well and therefore not an independent sample of it, does not win on the complementary 420, and still does not win on the 80 complementary rows that are *also* at 298 K, which is the block the temperature half of that claim rests on and the weakest of the three: $P \approx 0.3$ and no interval, and the three blocks were never tested against one another—and that block split is the whole of the evidence for the reversal being chemical. We claim only that **the fidelity lever we can demonstrate is confined to the smaller set's chemistry**, and that null is additionally a small-molecule and glycol-heavy one, so on drug-like solids the lever is untested rather than absent. Under the 2010 kernel's own typed latent, 306-dimensional and so several times worse resolved, the floor checks return two aggregates of opposite sign—+1.29 on the broad IDAC set ($P > 0.95$, $\approx$0.9, $\approx$0.9 over pair, two-way and source clusters) against -0.77 at $P \approx 0.3$ on the VT-2005-matched set, only the first of them standing—which is loss of resolution and not a demonstrated flip to the inputs, a flip needing a *lower* bound on B_{insuff}. The input-fixed control and its three limits, the three arms on both sets under both conventions (Table S8), and the third block at 298 K are in §S6.6.

Where B_{closure} dominates—on this set, the one row set the rule leaves standing—the ceiling is in the map *as it is deployed*: the fixed kernel together with the one computed conformer it reads, which is the composite B_{closure} is the error of (§2.1.1) and the object the pipeline runs on. Which of the two carries the error is a question this section does not answer; nor does dominance say the composite is *fixably* wrong. A small structure-preserving residual on the fixed COSMO-SAC exchange-energy kernel, fit and five-fold cross-validated on the 60 pairs of the VT-2005-matched set on their reference VT-2005 profiles, removes $46 \pm 5\%$ of the reference-input error under random folds but $-5 \pm 16\%$ once folds are grouped so that no solvent is shared, and only the grouped design is free of leakage, the closure error being predominantly a between-solvent systematic. It gives a null and not a bound: the $\pm$ prices only optimiser noise over five fit seeds, the held-out fold spread that dominates it going unpropagated (§S3). B_{closure} is therefore an error a handful of kernel parameters can absorb *within* a solvent and that we have not shown to be correctable across solvents. Where it fails is chemically placed, and placed differently on the two sets. On the VT-2005-matched set 90% of the squared error falls on strong hydrogen-bond-donor solvents and the closure is near-exact on acceptor-only ones—the chemistry COSMO-SAC's documented single-parameter H-bond term governs, the term the donor/acceptor-typed 2010 closure was built to replace. Half of that reading carries to the published evaluation of Table 2 and half does not, and on the broad set neither it nor its replacement reproduces, so we state neither (§S6.6). It places the error in a *chemistry* and not in one component of the composite: the reference profile for each of those solvents is one gas-phase conformer too, and neither set separates a displaced reference from the kernel.

That COSMO-SAC-2002 is a low-fidelity closure is the contribution rather than a qualification of the result: it is what a practitioner adopts by default, a closure's fidelity is *not knowable in advance* on a new system, and the field routinely composes an expressive encoder with a fixed mechanistic map and assumes the map is good enough. What follows when it is not is that the learned latent silently departs from the quantity it is named after (§3.3), invisibly without an external oracle. Outside that one row set the map reports cells it cannot resolve, not cells in which the closure is sound. One arithmetic sizes that exposure, and it is worth stating because the ground truth charging B_{closure} is quantum chemistry and not experiment: one closure read on two profile databases—HF-TZVP against BP-TZVPD-FINE—differs by a factor of 2.6 in AAD (Table 2), against a glycol-ether stratum, computed on the UD profiles the broad set is matched to, whose margin closes only under a more than ten-fold reduction in its squared error; so a profile source better than UD by the whole of that published bracket would leave that sign standing. The two sides therefore name different databases, and the bracket enters as the size of a database-to-database effect rather than as a comparison either side is on. The arithmetic holds the insufficiency bound fixed, which a new profile source would not, so it sizes the exposure and does not bound the share; the working, and the route by which a displaced reference loads onto B_{closure} through the curvature of g even where g is exactly specified, are in §S6.6 and §S9.

3.6 The same split on a second closure: pKa through a fixed Hammett relation

The instrument is not specific to COSMO-SAC, and a second domain runs it where the sign of grounding can be predicted before the measurement. The closure is a Hammett linear free-energy relationship $g(z^\star) = \mathrm{p}K_{a,0} - \rho\,\sigma^\star_{\mathrm{H}}$ in a substituent constant, with $\mathrm{p}K_{a,0}$ and ρ the tabulated Hansch–Leo–Taft reaction-series constants, one literature pair per scaffold—$(4.20, 1.00)$ benzoic acid, $(9.99, 2.23)$ phenol, $(4.60, 2.77)$ anilinium, $(5.25, 5.90)$ pyridinium—never fitted or refit on these data, so g is fixed in the sense Lemma 1 requires and the same g runs inside every split. It reads two things, so $z^\star$ is the pair $(s, \sigma^\star_{\mathrm{H}})$: the series s, which selects that literature pair, and the substituent constant at the tabulated value the rule below assigns. Only the constant is learned—the learned intermediate $\hat{\sigma}_{\mathrm{H}}$ is that constant predicted from structure; the external reference is the Hansch–Leo table,[31] printed in full in §S8; and the data are the OPERA experimental compilation[32] under a uniform, closure-independent quality filter that drops pre-ionized microstates.

The reference constant is assigned mechanically, by a rule that reads structure alone and never the residual (§S8), and that rule is also what defines the two poles. A record enters under one of the four reaction series—benzoic acid, phenol, anilinium, pyridinium—which fixes its $(\mathrm{p}K_{a,0}, \rho)$; $\sigma^\star_{\mathrm{H}}$ is then the sum, over the substituted ring positions, of the Hansch–Leo constant of that substituent *at that position*, σ_m meta and σ_p para, taken from a fixed list of nineteen. An ortho position carries no tabulated constant and enters the sum as zero, and so does a substituent outside the list. The first pole—the *clean* one below—is the 158 records whose every substituted position is meta or para with a tabulated substituent; the *degraded* pole is the 846 where at least one position is neither—571 of them carrying an ortho substituent, the other 275 an untabulated one—and there $\sigma^\star_{\mathrm{H}}$ is the partial sum over the positions it can see. The split is not benzenoid against heteroaromatic: all four series occur on both sides, 32 of the clean pole's records being pyridinium.

The two classic failure modes of that relation fall onto the two channels of Lemma 1: *resonance saturation*, a curvature the linear closure cannot represent, lands in B_{closure}, while what the zeroed positions carry—ortho field and steric effects, and whatever an untabulated substituent contributes—is information the intermediate lacks and lands in B_{insuff}. The two poles differ accordingly. On the clean one, the chemistry the relation was built for (n=158), it is well specified: RMSE 0.55, and the lower bound $\mathrm{MSE} - B^{\mathrm{up}}_{\mathrm{insuff}}$ on the non-negative B_{closure} is -0.2 (95% CI $[-0.4, -0.0]$)—below zero, where a lower bound sits when the closure predicts as well as a flexible regressor on its own argument—while the per-scaffold systematic offsets are ≤ 0.19 in $\mathrm{p}K_a$, whose square is the independent Jensen lower bound: ≤ 0.034, 11% of the 0.303 total. **That pole reads as well specified only after the quality filter**: the rule is closure-independent—net formal charge $\neq 0$, applied uniformly to both poles, never reading a residual—but it drops five pre-ionized microstates, and on the 163 rows before it the same floor is $+0.55$ rather than -0.2, i.e. positive and establishing $B_{\mathrm{closure}} > 0$ (§S8). On the degraded pole (n=846) it degrades to RMSE 2.26, of which the same estimator places at most 4.5 of the 5.1 total in input insufficiency and hence at least 0.6 in the closure—both figures one-sided, in the same direction as the solubility bounds. That pole's RMSE is essentially unchanged by the filter, so the filter does not manufacture the contrast between the two poles; what it decides is the sign of the clean pole's floor.

That estimator is not the eight-bin coarsening cell of §2.5, and the functional is not the same either: the two figures just given are floors $\mathrm{MSE} - B^{\mathrm{up}}_{\mathrm{insuff}}$ on B_{closure}, which when positive establish $B_{\mathrm{closure}} > 0$, and not the separation margin $\mathrm{MSE} - 2B^{\mathrm{up}}_{\mathrm{insuff}}$ of Table 1, which when positive would establish the stronger $B_{\mathrm{closure}} > B_{\mathrm{insuff}}$. Neither coordinate of $z^\star = (s, \sigma^\star_{\mathrm{H}})$ needs binning—s takes four values and $\sigma^\star_{\mathrm{H}}$ is a scalar—so the conditional variance is estimated directly, as the *out-of-fold* residual mean square of a random forest of $\mathrm{p}K_a$ on $\sigma^\star_{\mathrm{H}}$ fitted *within* each series, which exceeds B_{insuff} by the regressor's own approximation error and so leaves a lower figure on B_{closure}. Conditioning on s is not a free choice and it is not the conservative one: pool the four series and the same estimator returns 8.3 against the degraded pole's MSE of 5.1, a conditioning under which g is not measurable, so Lemma 1 does not license it and the closure floor it leaves would in any case be vacuous. The floor of 0.6 therefore rests on s belonging in $z^{\star}$, which is what the lemma demands of a closure that reads it; the fold design, and why it escapes the objection §3.5.1 raises against fitted estimators on solubility, are in §S8.

The decisive test is the same oracle swap, measured *in-distribution* across K=20 within-scaffold splits: the learned $\hat{\sigma}_{\mathrm{H}}$ replaced by the tabulated $\sigma^{\star}_{\mathrm{H}}$ in the same closure—the series label is read from the structure in both arms, so the swap moves one coordinate of $z^{\star}$—the sign of the change recorded per stratum. It flips. On the clean pole (n=158) the reference *improves* accuracy—MAE 0.48 learned against 0.315 with the reference, a gap of -0.165 at a 90% CI of $[-0.28, -0.04]$, better in 19 of 20 splits and in three of the four reaction series—so the ceiling there is the noisy learned estimate and not the closure, which is an internal positive control on real data. That control is the filtered 158-record pole's and inherits its scope: on the unfiltered 163 the closure floor is positive, and "well specified" is then not available to explain the swap's sign. On the degraded pole (n=846) the same swap *degrades* it—0.83 against 1.62, worse in 20 of 20 splits and in all four series, a gap of $+0.79$ at a 90% CI of $[+0.57, +0.95]$—the freely learned constant absorbing what the zeroed positions carry and the tabulated one lacks. The sign of grounding therefore reverses inside one closure, in the direction Proposition S1 gives a priori and—consistently with the one-sided bound above, though not isolated by it—through the insufficiency channel rather than the closure-error cancellation of §3.3. Both signs are one comparison: the sign of the grounding gap is by construction that of the learned arm's error against the fidelity-set oracle error, and what a competence sweep adds is that the oracle side of it does not move with the training budget while the learned side does. The clean pole's threshold ($\approx$ 0.315) is never beaten by the learned arm, so grounding helps at all four budgets swept, while the degraded pole's threshold (1.62) is high enough that the learned arm beats it across the whole interpolation range, down to 12% of the data, and grounding reverts to helping only under scaffold *extrapolation*, where the learned constant breaks (2.6±2.0) back above the threshold—in the mean over the six splits, three of which still have grounding hurting. The clustering unit is the Hammett reaction series, of which only G=4 exist, so the bootstrap resolves nothing finer than series-level sign agreement, read as a sign test on four units: p=1/16 on the degraded pole, where grounding hurts and all four series agree, and $p \approx 0.31$ on the clean one, where three of them do—so that test carries the degraded pole and not the positive control, which rests on its splits and its gap instead. The construction, the estimator validation and the measured fidelity sweep are in §S7.

3.7 Scope, accuracy, and limitations

The result is a cautionary measurement about one of the two operations the word "grounding" names, and it is two measurements made in different places. On solubility: supervising a learned physical latent on reference values during training helps, and *supplying* those values at inference hurts. On infinite-dilution activity data, and there on one class of solvent: the fixed map's own error stands above the bound on what its inputs fail to carry—the condition under which better inputs surface a map's bias rather than help, and one the standard route to a better closure did not clear either. The two are measured in different regimes on largely disjoint chemistry, no measurement here connects them, and what would settle it is named in limitation (i) below and in (x); until it is run, "the substitution hurts *because* the map is misspecified" is an inference and this paper does not assert it. Both sets are scoped by the reference database that defines them and neither is the deployment regime. The useful form of the finding is one-sided and narrow: a place where grounding effort is *not* the remedy, on one row set of the map, rather than an allocation across chemistries.

The unconstrained control's representation. If the physics arm's intermediate is a surrogate, the thermodynamics might still be present implicitly in the control, which is free to represent it. It is not. A ridge probe of the control's pair representation recovers the model's own prediction at $R^2 = 0.98$ but not the measured ideal-solubility term. It is scored on 13 held-out solutes / 527 rows at zero solute overlap, and that term is 92.2% between-solute variance, so the effective n is 13 clusters and not 527 rows: $R^2 = -0.506/+0.111/+0.027$ at seeds 42/43/44, permutation $p = 0.894/0.055/0.177$ against a 1000-draw null, and a between-seed range of 0.616 (sd 0.334) larger than any one of the three values, so a single seed would have been unreadable and seed 43 alone would have read as a near-miss. The null itself reaches $R^2 \approx 0.32$–0.38 at this sample size, so what the negative supports is an upper bound at $R^2 \lesssim 0.30$ rather than an exclusion (limitation (ix)). What the representation is organised by instead is scaffold identity, whose share of the representation's variance sits at the solute-identity ceiling—on a split where those two partitions nearly coincide by construction, only 11 of the 124 test scaffolds holding more than one solute, so they agree on 91% of the data and a share at that ceiling is weak evidence of within-scaffold indistinguishability rather than a measurement of it. A test molecule on an unseen scaffold is not a small extrapolation but a point off the axis the representation is organised along. That is a description of this representation and not a diagnosis of the encoder, which the linear probe of §S6.1 does not supply. The full tables, the controls, the direct within-scaffold measure on the solutes that do share one, and the temperature probe are in §S10.

The contribution. The unit of contribution is the measurement, not any single number it returned. Its two ingredients are individually standard: the closure–insufficiency split is a law-of-total-variance identity, and the sign of grounding's effect follows the model-discrepancy principle.[19,20] Their combination is an instrument—an identity on a *fixed* operator, with the insufficiency side bounded rather than point-identified, and a sign rule testable *a priori* and in both directions. The pKa domain supplies that test as an in-distribution sign flip, an oracle-on-a-fixed-operator affordance the concept-leakage setting lacks, its intermediate having no ground truth.[17,18] The solubility results are the instrument's first application. Both scopes are narrow: the sign flip resolves no finer than the four reaction series it is read on, and the solubility results' transfer out of the matched regime is not established.

Where the accuracy stands, and the confound in it. Table 3 places every arm of this work beside the external models, in two blocks that may not be read across. The physics arms trail their own control in the seed mean—the COSMO-SAC arms at every seed, the NRTL arm at two of three, being ahead at seed 42 (1.734 against 1.749; Table S3)—and the predict-the-mean row shows how little room sits above them. The physics and control arms were tuned separately and no hyperparameter-matched control was run, so the global ΔMAE of +0.14 between them on the 5608 labelled test rows—from the unrounded means, where Table 3's rounded cells differ by 0.15—carries no information about the thermodynamic bottleneck, and neither do the per-solvent-class map, the per-solute-class map, the novelty inversion or the data-efficiency trend, all of which resolve the *same* difference along chemical axes. We claim no accuracy result in either direction; those strata are reported in §S6.4 and §S6.5 as descriptions of where two particular pipelines differ, and no conclusion rests on them. Running the matched control is the direct repair and we have not run it. What survives the confound is every *within-checkpoint* contrast, where the two sides share one trained model and one tuning: §3.1, §3.2 (the ranking result included), §3.3 with the output correction folded into it, and §3.5, which uses no trained model at all. Separately, and not as an accuracy claim about the bottleneck: no configuration we ran attains a lower *seed-mean* error than the control—at one seed the NRTL arm does—and our one attempt to move the ceiling by grounding the crystal factor on external labels tested nothing, its pool being 97.76% melting points the model was already trained on (§S6.3). The one place the physical *structure* helps is temperature extrapolation, and it helps there as a property of the hypothesis class. Whether a trained arm inherits it we do not know: the only checkpoint ever scored on the temperature-withheld split is one whose predictions are near-constant, so it is untested rather than answered (§S6.1, §S5).

The external block is not like-for-like and the table says so in its own block heads. FastSolv and SolProp are deposited here on a *by-solute* split, this work's arms on the harder solute-scaffold split, and the published figures are as published, on other corpora and in another unit; retraining the two on the scaffold split needs a GPU run we did not perform, and scoring released FastSolv weights on our test rows is invalid because FastSolv is trained on this corpus. One sentence survives all three caveats, and it is not a favourable one: this work's control, at $\log_{10} S$ RMSE 1.00 on held-out scaffolds, is the same order as published new-solute extrapolation—the published leaders run 0.83 to 0.99 in that unit and the physics-informed cycle 1.43 to 2.16—rather than an order away, and the physics arms, at 1.39 to 1.60, are worse than every published leader at every seed and every clip. What may be read from those figures is what survives the clip sweep and not the two decimals; the sweep, the published rows themselves and the full reading are in §S4.

The sign rule's own prediction. Proposition S1 predicts *when* to expect these negatives: a prior hurts precisely in the mature-data, low-noise, well-covered regime. The synthetic sweep (Fig. S9) is true by construction there, so it verifies the instrument rather than the hypothesis, while the pKa closure exercises the sufficiency channel on real data. Whether grounding helps is shaped by *two* axes, closure fidelity and how the latent is trained; that plane is not characterised on real systems here. The chemical strata form a pattern in the direction the sign rule would give, but separately tuned pipelines can produce it and there is no multiplicity control, so §S6.4 records it as the experiment a matched control should run and the diagnostic is *proposed* rather than validated.

Limitations: what the sets can carry. (i) The phenomenon and its causal attribution lie in *different regimes*: the paradox is measured where the model operates (finite composition, all γ, n=5608), the attribution and the latent-departure result only on infinite-dilution activity data (n=477 across temperatures, and the 298 K, low-γ, VT-2005-matched n=60 and n=44 sets). Transfer back to the paradox's own regime is a conjecture and is the principal open question here. The n=60 set is narrowly sourced—all by gas chromatography and all from three publications, 33 pairs from one—so neither a method-stratified nor a source-clustered re-estimate is available on it, and its margin is extraction-sensitive: a broader ThermoML pull at 298 K widens it but *inverts* it on the gas-chromatography subset, a sensitivity we leave unresolved. The broad set does not move that way under the same cut and is not the set the paradox's own oracle is defined on; the arithmetic and the enlarged-pull values are in §S9.

Limitations: what the estimator can deliver. (iv) B_{insuff} is upper-bounded, not point-identified (no replicates at fixed $z^\star$); the smoothness-free Jensen bound assumes m and g share the $\ln \gamma_2^\infty$ reference state, established by the closure validation of §2.3, so the mean offset is decoder bias and not a units artifact. B_{closure} is the error of a composite—the fixed map together with the calculation that produced $z^\star$—and no measurement here divides it between the two; §3.5.3 sizes that exposure and leaves the share unbounded. (vi) The stratification reads structure alone, so no cell was selected on its outcome, and it is exhaustive and disjoint *within* an axis but not across the two axes or their granularities: 362 of the 1711 stratum pairs share rows, all 37 water-as-solute rows lie inside three solvent classes, and 108 of the 129 alkane-solute rows are glycol-ether rows, so cells from different axes are not independent evidence and those rows are counted once. Granularity is an analyst choice, which

Table 3: Solubility accuracy in two blocks that may not be read across: this work on the solute-scaffold split, external models re-run here on an easier by-solute split. Two sample sizes appear where a row's two metric columns are not scored on the same rows. The published figures, the random-pair rows, and the conversion and saturation clip behind the $\log_{10} S$ column are in Table S17.

Model	n ($\ln x_2/\log_{10} S$)	MAE $\ln x_2$	RMSE $\log_{10} S$	R^2 $\ln x_2$
This work, solute-scaffold split; mean ± s.d. over seeds, three unless the row says otherwise				
DirectGNN, same encoder, no closure *(control)*	5608 / 5440	1.70 ± 0.03	1.00 ± 0.02	0.42 ± 0.03
TGNN, NRTL closure (physics)	5608 / 5440	1.79 ± 0.07	1.54 ± 0.07	0.34 ± 0.03
TGNN, COSMO-SAC closure, supervised $\hat{\sigma}$ (physics)	5608 / 5440	1.85 ± 0.05	1.39 ± 0.19	0.33 ± 0.05
TGNN, COSMO-SAC, reference $\sigma^\star$ substituted at evaluation	5608 / 5440	2.25 ± 0.02	1.60 ± 0.14	-0.03 ± 0.04
Scale reference on the same solute-scaffold rows				
predict the test-set mean of $\ln x_2$	5608 / 5440	2.32	1.33	0.00
External models re-run on this corpus on a by-solute split, one run each—not a ranking against the block above				
FastSolv, retrained on this corpus, re-scored	10287	1.44	0.84	0.55
SolProp cycle, released model + 3-parameter recalibration	10287	2.34	1.34	−0.18
SolProp cycle, released model, zero-shot	10287	2.36	1.87	−0.47
SolProp encoder retrained on $\ln x_2$, cycle discarded	10570 / 10287	1.62	1.07	0.39

is why three solvent levels and two solute levels are all reported. There is no multiplicity control on the per-cell bootstrap frequencies, so one cell's P describes that cell alone, and the admissibility rule is a stability filter and not a correction for the search; six of the nine solvent classes are not boundable here and three carry no estimate at all. What the search survives is measured rather than assumed, and neither test establishes the finding as a chemical effect: under the chemistry-blind relabelling null of §3.5.1 the shape of the finding actually claimed—the number of admissible *positive row sets*, and the certified magnitude—is not above chance, and the pre-declared out-of-sample test of the chemistry returned *not testable*. The single finding is therefore stated at the level the instrument licenses on its own rows and *not* as a multiplicity-controlled chemical effect, and the same holds a fortiori for every cell the map reports and does not state. The rule is also one-sided in what it can deliver: with B_{insuff} bounded only from above, no cell can ever return “better inputs would help here”, so the map only ever withholds a chemistry from an allocation of grounding effort.

Limitations: what is untested, and what is not auditable. (xii) The scaffold-leak guard is a property of the released stream builder and not a certified property of these runs, so the 0.20 supervision gain is the one reported number a leak could inflate and is uncertified until the leak-free re-run (§2.2); the arguments that bound the other contrasts do not restore auditability. The nine remaining limitations are in §S6.7, keeping their numerals: (ii) what the convention rule costs the aggregate and (iii) which other cells of the map keep their sign, both already stated where they bite, in §2.5 and §3.5.1; (v) the constant-offset bound's convention-specificity, likewise in §2.5; (vii) the accuracy ceiling against the noise floor, (viii) what the three-seed latent-departure result establishes and what it does not, (ix) the reference scarcity that bounds the black-box probe, (x) the closure-fidelity contrast and the two runs that would settle it, (xi) the separate-tuning confound stated above, and (xiii) what is checkable only by re-running the released code.

4 Conclusions

Grounding a learned physical intermediate in reference values is two operations, not one, and here they had opposite signs. Training the intermediate toward its reference improved the model, on a contrast this work does not certify leak-free; supplying that reference at prediction time made the model worse, at every seed, and no question of certification reaches that second half. The operation that failed is the one a practitioner reaches for, because it is the one that needs no retraining. The two terms of the error can be told apart on activity data and not on solubility, and there, on one class of solvent, the fixed thermodynamic model's own error—the model as deployed, reading one computed conformer per molecule, and not its kernel alone—stands above the bound on what its inputs fail to carry: the condition under which a more accurate input exposes a model's bias instead of removing it. That is the reading the failed substitution invites, measured on other data and not a test of it. It is also where the instrument answered rather than where the model is shown to fail, on a row set that survived a search over every stratum, is not established against that search, and for which the one independent database supplies no retest; and the answer came one chemistry at a time: a single figure for a whole data set is an average over that set's composition, a property of the set and not of the model. The intermediate paid for the arrangement. Trained through a model it cannot correct, and measured on runs of its own, it stopped reporting the physical quantity it is named after wherever a reference existed to hold it against—which is the price of reading a grounded latent as a measurement.

The general statement is a design rule with a test attached. Wherever a predictor composes a fixed physical model over a learned intermediate for which an external reference exists, that reference measures which of the two limits accuracy, and where it answers, the answer decides the remedy: a more faithful physical model—the calculation that supplies its reference values included—or better inputs. The answer comes chemistry by chemistry rather than once for a field, and it comes one-sided, so what the measurement returns is the chemistry on which better inputs are not the remedy—here one class of solvent, on which the standard route to a more faithful model did not clear the bar either. What this work establishes about the instrument is the map itself: which cells it can answer in, and at what size. Whether a latent trained through an imperfect model becomes, in general, a surrogate for that model's error is the prediction this work leaves testable.

Author contributions

N. L. Polomoshnov: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. A. V. Rudik: Writing – review & editing.

Conflicts of interest

There are no conflicts to declare.

Data availability

The analysis code, and those intermediate artifacts small enough to version, are available at `https://github.com/doctawho42/tgnn-solv` (commit **[hash to be supplied at submission]**) under the MIT licence, that commit being the one the reported numbers were produced from; the model checkpoints, the processed split, the full per-arm prediction files and the larger analysis artifacts will be deposited in a Zenodo archive, **[whose DOI is registered at submission]**. The data sources are public: BigSolDB 2.0[23] for solubility, the VT-2005 database for the reference σ-profiles, Ref.[33] as redistributed with Ref.[25] for the UD profiles, ThermoML for the infinite-dilution activity coefficients, the PGL 6th-edition IDAC database for Table 2, and the OPERA compilation[32] for the pKa data. Four disclosed defects sit on the deposited record, in §S1, §S9 and §S3: the three-seed surrogate run's per-run split provenance was not logged, one σ-stream build on disk was written against the wrong split files, the two VT-2005-matched audit artifacts disagree on one out-of-fold estimate, and the COSMO-SAC-2010 layer's reference reproduction (§2.3) was printed to its validator's console and never written to an artifact at all. Two further defects are repaired rather than disclosed, each described where its numbers are printed: the two FastSolv retraining bundles were re-scored (Tables 3 and S17, with the diagnosis, the withdrawn values and the control that reproduces them in §S4) and the withdrawn bundle is kept beside the re-scored one, and the truncated seed-44 σ-oracle per-row file has been recovered complete and now stands in the deposit (Table S25, §S9).

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Supporting Information Available

The Supporting Information is a separate document. Its sections, tables, figures and equations are numbered S1, S2, …, and every reference above that carries an S is to it. It contains supplementary methods; the remaining proofs; per-arm tables and decomposition robustness; the map's 59 strata in full, including the 44 the estimator cannot bound; further external ablations; supplementary mechanism results; supplementary diagnostics and broad negatives; the second-domain pKa construction; data provenance and reproducibility; the black-box probe tables; the synthetic closure-fidelity family; and the positioning against adjacent literatures. Two machine-readable tables are deposited with it: the 477-row broad IDAC set (Data Set S1, `broad_idac_set_477.csv`) and the 60-row VT-2005-matched set (Data Set S2, `vt2005_matched_set_60.csv`); their columns are listed in §S9.

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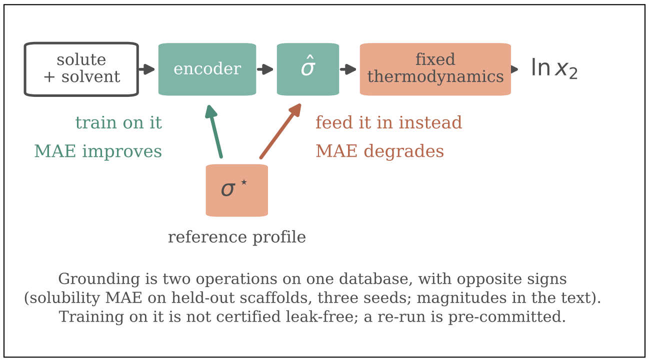
TOC Graphic


solute
+ solvent
encoder
$\hat{\sigma}$
fixed
thermodynamics
$\ln x_2$
train on it
MAE improves
feed it in instead
MAE degrades
$\sigma^\star$
reference profile
Grounding is two operations on one database, with opposite signs
(solubility MAE on held-out scaffolds, three seeds; magnitudes in the text).
Training on it is not certified leak-free; a re-run is pre-committed.